Problem Workbook

Real Estate Finance and Investment

Eighth Edition

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PREFACE

This Problem Workbook supplements Clauretie and Sirmans, *Real Estate Finance and Investment, 8th Edition*. The purpose of the workbook is to reinforce the learning process by aiding in your understanding of the concepts presented in the text. Much of real estate finance is understanding the structure and pricing of mortgages and mortgage-related securities. Even if your exposure to the mortgage market is limited to simply financing your "dream home", understanding the process can be critical in making the optimal financing decision.

For appropriate chapters the workbook contains practice problems designed to help you better understand the theory and concepts. Not all chapters are problem-oriented thus the workbook does not contain problems for all chapters. However, the workbook is organized by chapter and, where appropriate, the type of problem is indicated.

APPENDIX ONE
A REFRESHER COURSE ON THE TIME VALUE OF MONEY AND RELATED CONCEPTS

Understanding the concept of time value of money is critical to understanding the workings of the mortgage markets and the structure of mortgage instruments and mortgage-related securities. As a review, this section contains time value of money problems. These problems coincide with Appendix 1 and are designed to provide practice on the general time value of money concepts.

A.1. Old Bank offers to pay 6 percent interest, compounded annually. New Bank, to be competitive, offers 6 percent interest, compounded monthly. If you buy from each bank a five-year, $1,000 certificate of deposit, with all interested compounded, what is the difference in values at the end of five years?

A.2. Mr. Ace Investor has $10,000 to invest for one year. He has three investment alternatives: (1) earn 8 percent, compounded annually, (2) earn interest compounded quarterly, and (3) earn interest compounded monthly. What must the nominal interest rates be on the second and third options to make all the investments earn the same yield?

A.3. An apartment house has a projected net income of $15,000 per year, and its projected net sales price after five years is $150,000. Considering its risk, you require a 14 annual percent return on this investment. How much would you be willing to pay for it?

A.4. Jane Ire is offered a real estate investment that promises to pay $80,000 after 5 years. She feels, based on the investment's riskiness, that the annual rate of return should be 15 percent, compounded quarterly. What price should she pay for the property?

A.5. Peter Piper is offered an investment that will pay $5,000 at the end of each year for the next 10 years. He wants to earn an annual rate of return of 16 percent. How much is he willing to pay for the investment today?

A.6. Jim pays $10,000 for a mortgage contract that will pay $3,000 at the end of each of the next 5 years.

A. What rate of return will he earn?
B. Suppose that the annuity is paid as an annuity due. What is the rate of return in this case?

A.7. The value of a house today is $98,000. If it has increased in value at 6% per year, what was the value eight years ago?

A.8. Doctor John purchased 50 acres of land ten years ago for $800 per acre. A. If he could have alternatively invested the money at 8 percent per year, what price must he receive today to break even with his opportunity rate? B. If Doctor John sold the land for $1,400 per acre today, what was his actual rate of return?

A.9. Suppose you are interested in buying 25 acres of land to start a blueberry farm. The owner is willing to finance 70 percent of the $100,000 purchase price at 10% interest over 8 years.

A. What will be the payment assuming annual amortization?
B. What would the payment be if monthly payments are required?

A.10. The Smiths desire to purchase a house and they open a savings account that pays 5.75 percent interest, with monthly compounding. If they put $120 per month beginning one month from now and they must have 20 percent of the price of a house as a down payment, what price can they pay for a house after saving for 5 years?

A.11. Mr. Wrinter purchases 10 duplexes (20 rentable units) and he expects to have to replace the air conditioning

equipment in each rentable unit 10 years from now at a cost of $5,000 per unit. The bank that loaned Mr. Wrinter the money to purchase the duplexes requires that he deposit a part of the monthly rents into an account earning 6 percent interest to insure that the money will be available to replace the air conditioning units. How much money must Mr. Wrinter deposit at the end of each month?

A.12. You are considering the purchase of 75 acres of land that you believe will be developed as a shopping center. You estimate that you can sell the land three years from now at $25,000 per acre. How much should you pay now for the land if the required rate of return is 25 percent?

A.13. What is the current value of a leasehold interest where lease payments are $1,500 per month (annuity due) for five years? Assume a 10% discount rate.

A.14. Suppose that, as a lessee, you just signed a lease to pay $1,000 per month (annuity due) for 5 years. Simultaneously, you can sublease the property for $1,800 per month (annuity due) for 5 years. What is the value of your leasehold interest at a discount rate of 12%?

A.15. What is the present worth of an income-producing property which receives an after-tax cash flow of $20,000 in year one, $22,000 in year two, $25,000 in year three, $30,000 in year four, and $32,000 in year five? Assume the discount rate is 15 percent.

A.16. Tom makes a deposit of $5,000 in a bank account that earns interest annually at the rate of 8 percent.

A. How much will Tom have in the account at the end of 5 years?
B. Assuming the account earns 8 percent compounded quarterly, how much will he have at the end of 5 years?
C. In comparing A and B, how much additional interest do you earn with quarterly compounding?
D. What are the effective annual yields for each alternative?

A.17. Which option would you prefer: A 1 year investment of $50,000 earning 6 percent interest compounded monthly or a 1 year $50,000 investment at 7 percent compounded annually?

A.18. Mike is considering the purchase of a vacant lot. He expects the price of the lot to be $30,000 at the end of 8 years. He requires an investment yield of 10 percent annually.

A. Should he buy the lot at the asking price of $15,000?
B. What is the effective yield of the investment if Mike does purchase the land for $15,000 and sells it 8 years later for $30,000?

A.19. You can buy land that you expect to be worth $80,000 after 4 years. You require a 14 percent annual return. How much should you pay for it today?

A.20. Harry plans to deposit $4,000 at the end of each 6-month period for the next 10 years. He can earn annual interest of 6 percent.

A. What is the balance at the end of 10 years?
B. If the deposits are structured as an annuity due what is the balance at the end of 10 years?

A.21. Suppose you deposit $2,000 at the end of each month in an account that earns a10 percent rate, compounded monthly. How much will you have at the end of 6 years?

A.22. Suppose you deposit $1,000 at the end of year one, nothing at the end of years two and three, $2,000 at the end of year four, and $500 at the end of year five. Assuming that interest is compounded annually at 6 percent, how much will you have in the account at the end of 10 years?

A.23. Mary is considering an investment that will pay her $5,000 at the end of each year for the next 10 years. She requires an annual return of 14 percent on this investment. What is the maximum that she can pay today for the investment and earn her required return?

A.24. As the owner of an apartment complex, you know that you will have to spend $50,000 on air conditioning units 10 years from now. A. If you can earn 8 percent on deposited funds, how much must you deposit at the end of each year to accumulate the needed funds? B. If you are willing to make a deposit every six months (ordinary annuity), how much must the deposit be to accumulate the funds?

A.25. You have purchased a building for $1,000,000 and expect it to grow in value at an annual rate of 7%. What will the building be worth in ten years?

A.26. If the building in question 1 is, in fact, worth $2,000,000 at the end of ten years (and there were no other cash flows) what annual rate of return would you have received on your investment?

A.27. If, in addition to being worth $2,000,000 at the end of ten years it also provided cash flows of $30,000 annually, beginning at the end of the first and ending with the end of the tenth year, what rate of return would you get on your investment?

A.28. You purchase a building for $1,000,000 and sell it for $1,500,000 at the end of five years. What annual (equal) end-of-year cash flows would also have to be forthcoming from the investment in order to make a 12% annual rate of return?

A.29. What is the present value of a lease agreement that requires you to pay an annual lease of $40,000 for five years beginning at the end of the first year if your discount rate is 12%?

A.30. What would be the present value of this same lease if the first payment were to be made at the beginning of the first year?

A.31. What is the present value of a lease that requires you to pay $40,000 at the end of the first year and then escalate the payment by 4% annually for the remaining four years (five yearly payments in all) if your discount rate is 13%?

A.32. You are a property owner who leases a property for $50,000 annually beginning at the end of the first year for ten years (ten payments) You take each of the lease payments and invest them in a bank account that pays 8% annually. How much money will you have in your bank account at the end of the tenth year?

A.33. You can be the lessee of an office facility (lease A) for five annual annuity due payments of $55,000. You can lease an identical facility (lease B) for $45,000 annually for five years but must spend $35,000 in tenant improvements at the beginning of the first year. If your discount rate is 12% which of the two leases should you accept?

A.34. You plan to retire at the end of ten years. You want to have a fund at that time that will allow you to draw out nine annual payments of $100,000 beginning at the end of the first year of retirement. You can make 9% annually on the fund at all times. What is the amount of ten equal annual payments you must put into the fund (the last payment on the date of retirement) if you want to accomplish your goal?

A.35. In question A.34 what would be the amount of the annual payment if you made only 6% on the fund while putting the money in but made 9% at the time of endowment (retirement)?

A.36. If you can make 7% annual return on an investment how long will it take you to double the value of the investment?

A.37. If you double the value of an investment in eight years what annual rate of return do you get?

A.38. If you take out a mortgage for $100,000 with an annual payment for 30 years at 9% annual rate what will be payment if the first payment is made at the end of the first year?

A.39. You have $5,000 in a bank account that earns 8% annual interest. You want to accumulate $45,000 by the end of the tenth year. What is the amount of the annual payment into the account that you will have to make at the end of each of the next ten years to accomplish your goal?

A.40. You have $10,000 in a bank account that makes 8 percent annually. At the end of the fifth year you withdraw $8,000 and place that in an account that makes 9 per cent annually. The original account continues to make 8 per cent annually. How much money will you have in the two accounts at the end of the ninth year?

CHAPTER 1
REAL ESTATE FINANCE: AN OVERVIEW

This chapter provides an overview of real estate finance. It divides the field of finance into sub-fields of which real estate finance is but one. The environment of real estate finance is discussed. You should have a basic understanding of credit flows and the function of financial intermediaries. You should also understand that real estate finance deals more with the capital markets than the money markets.

Mainstream finance studies the process and the role of the financial markets in the transfer of funds between institutions, individuals, and government. The primary focus is wealth-maximization. Concepts such as time value of money and techniques such as discounted cash flow play important roles in achieving the goal of maximizing wealth.

Real estate finance is a much narrower body of study with the focus on the mortgage markets and the financing of real estate. Examples of study topics include the valuation of mortgages, the effect of financial leverage on the value of real estate, the primary and secondary mortgage markets, and the securitization of mortgages. While the primary mortgage market plays the critical role of mortgage origination, the secondary mortgage market plays an equally-critical role in providing funds to the primary market through mortgage securitization.

CHAPTER 2
MONEY, CREDIT, AND THE DETERMINATION OF INTEREST RATES

This chapter acquaints you with the fundamental relationships between monetary policy, monetary growth rates, inflation, and interest rates. You should have a basic understanding of how interest rates are determined and the role of inflationary expectations. You should also understand how the individual characteristics of credit instruments affect their yield. The following problems deal with the fisher equation, the equation of exchange, and yield curves.

2.1. Assuming that the Fisher Equation is accurate, provide the missing data in the following table.

Scenario	Nominal Interest Rate	Real Interest Rate	Expected Inflation
A	10%	4%	
B	9%		4%
C		3%	3%
D	8%		5%

2.2. Using the Equation of Exchange, fill in the missing values in the table.

Scenario	Money Supply	Velocity	Index of Price Level	Real Transactions	GNP
A		5		750B	7500B
B		4	8	800B	
C	500B		10		3000B
D	500B	3	5	300B	
E		5	6		3000B

2.3. If the velocity of money remains constant in the Equation of Exchange, what will the rate of inflation be under the following conditions:

Current money supply = $2,000B
Velocity of circulation = 6
Real transactions = $3,000B
Price level index = 4
Next year's money supply = $2,400B
Next year's real transactions = $3,200B

2.4. Fill in the missing values in the following table.

Scenario	Nominal Rate	Risk Free Rate	Default Risk Premium	Callability Risk Premium	Maturity Risk Premium
A	10%	4%		2%	2%
B		3%	3.50%	1.50%	3%
C	9.50%	4%		3%	2%
D	8%		3%	1.50%	1%
E	12%	4%	3%	3%	

2.5. If the current rate on a one-year bond is 6% and the current rate on a two-year bond is 4%, what is the expected rate on a one-year bond one year from now?

2.6. If the current rate on a one-year bond is 8%, the current rate on a two-year bond is 12%, and the current rate on a three-year bond is 15%, what is the expected rate on a one-year bond at the end of year one? What is the expected rate on a one-year bond at the end of year two?

2.7. Assume that the current yield curve appears as follows:

Maturity	Yield
1-year	5.0%
2-year	5.5%
3-year	6.0%
4-year	6.3%
5-year	6.4%

What are the expected one-year rate of interests for the following periods?

A. End of year one to end of year two.
B. End of year two to end of year three
C. End of year three to end of year four
D. End of year four to end of year five

2.8. Assume the yield curve is downward sloping for the first few years as follows:

One-year	6%
Two year	5%
Three-year	4.5%

What is the expected interest rate for a one-year security beginning at the end of year two?

2.9. If the expected rate of inflation for the next year is five percent and the real rate of interest is two percent, what will the nominal rate of interest be on a one-year default free security today if the Fisher Equation is correct?

2.10. The one-year default free rate of interest is 7%. The market anticipates a rate of inflation over the next year equal to 4%. What does this imply the real rate of interest is if the Fisher Equation is correct?

2.11. The current money supply is $500 billion and the velocity of circulation is four. Given that real transactions are $1,000 billion what does this imply about the price level if the equation of exchange holds?

2.12. If real transactions can increase four percent next year, the velocity of money remains the same but the money supply increases by ten percent give the new price level assuming the data in question 2.11.

2.13. What does this imply about the rate of inflation?

CHAPTER 3
FINANCE THEORY AND REAL ESTATE

This chapter helps you understand the important principles of finance theory and their relationship to real estate. These concepts include asset valuation, financial leverage, capital structure, option valuation, portfolio theory, and agency relationships. You should understand that the conceptual framework of finance can be applied to real estate. Thus, real estate finance is not simply a description of markets and institutions, but also involves economic models and financial economics.

When investors are contemplating investment alternatives, the primary focus is generally on the rate of return. However, risk plays an equally important role in investment decisions. In an environment of risk averse investors, there will always be a positive tradeoff between return and risk. To earn any return above that for postponed consumption, the investor must be willing to take on some level of risk.

Risk plays a role in the valuation of real estate since the projected cash flows from a project may have some level of riskiness and the discount rate used to capitalize those cash flows will reflect the perceived riskiness. Some of the riskiness of the cash flows may result from the use of debt financing in acquiring the asset.

When debt is used in the purchase of real estate, a mortgage is created. Mortgage contracts are somewhat favorable to borrowers in the sense that there are usually two options embedded in the contract: a call option and a put option. The call option allows the borrower (the issuer of the debt) to "call in" or redeem the debt any time before maturity. Thus, if a borrower originates a thirty-year mortgage but then wants to repay it in full at, say, the end of year five, the borrower simply exercises his/her call option. Even a prepayment penalty does not prevent the borrower from exercising the call option. The put option allows the borrower to "put" the property in place of the debt. The borrower does this by defaulting on the mortgage. Since the property is specifically named as collateral in the debt, the borrower puts the property in place of the debt by defaulting on the mortgage and thereby allowing the lender to sell the property to repay the debt.

Participants in both the real estate and mortgage markets are concerned with market efficiency. The primary question is: do observed transaction prices reflect the true value of the asset? Another angle to the issue is the question: how efficiently does the current price of an asset reflect the information available about the asset? In general, studies show that asset prices in most markets (real estate and mortgage markets included) reflect all publicly available information about an asset. History also shows us that market participants that have access to "inside" information can "beat the market" by earning excess returns.

Aligning owner and management interests is an age-old problem. Owners are constantly seeking ways to force or incentivize managers to act in the owners' best interest and not the manager's best interest. There are various ways to influence manager behavior. These include monitoring programs, bonding procedures, and structuring programs (compensation in stock options rather than salary, for example).

CHAPTER 4
RESIDENTIAL MORTGAGE FINANCING IN THE UNITED STATES

This chapter provides a history of residential finance beginning with ancient times to the present. The chapter explains how Roman, German, and English law are infused in our system of rules and regulations. The chapter gives the early history of residential finance up to World War Two. Then the chapter gives a decade-by-decade review of the transition of the residential finance market from the end of World War Two through the 2000s. Some important developments occurring over this period were maturity mismatch, disintermediation, creation of the secondary mortgage market, creative financing, and alternative mortgage instruments. Dramatic changes occurred in the residential mortgage market. Of particular importance was the role of interest rate volatility and inflation. You will also have a better understanding of the widespread failure of savings and loan associations in the 1980s caused by negative equity, maturity mismatch, and other factors.

The 1940s saw the creation of the Veteran's Administration with its housing program for WW II veterans. This program, along with the invention of the modern subdivision, became an integral part of the housing boom to meet the demands of returning GIs that occurred in the late 1940s and early 1950s. Following the 1950s expansion of the savings and loan industry, the 1960s saw the beginnings of severe future problems. These problems included crippling inflation, disintermediation (due primarily to regulation), and maturity mismatch. A positive note of the 1960s was the expansion of the secondary mortgage market and the creation of Ginnie Mae in 1968.

Freddie Mac, a major player in the secondary mortgage market was created in 1970. The 1970s saw increasing market interest rates due to inflation. By the end of the decade, mortgage lenders were pleading for an alternative to the standard fixed-rate mortgage. At this point, regulators allowed lenders to start writing alternative mortgage instruments such as adjustable-rate mortgages. This allowed lenders to shift at least some of the interest rate risk to borrowers.

The 1980s was a decade of regulatory and tax law changes, among other things. Tax acts created both incentives (Tax Act of 1981) and disincentives (Tax Act of 1986) to invest in real estate. Massive losses in the savings and loan crisis prompted a restructuring of the mortgage lending industry.

As inflation subsided in the 1990s, mortgage rates decreased. These relatively low rates prompted a refinancing craze that began in 1993 and ran through the remainder of the decade. Housing construction consisted primarily of up-scale housing to meet demand of baby boomers. Expansion of the world-wide web eliminated "local" shopping for mortgages and made mortgage lending national in scope.

Beginning in 2000, single-family homes entered the speculative investment market for the first time. Increasing demand compared to a relatively fixed supply resulted in rapidly increasing house prices. At their peak in 2006, house prices were up an average of almost 90% from their 2000 level. Due to affordability problems and for other reasons, mortgage lending standards were relaxed and a wide variety of mortgage loans were written. This prompted massive mortgage defaults beginning in 2007 and led to bailouts for secondary mortgage market participants such as Fannie Mae, Freddie Mac, and Wall Street. The end of the 2000s decade found house prices beginning to rebound and mortgage interest rates at an all-time low.

CHAPTER 4 APPENDIX A
THE ERA OF CREATIVE FINANCING

This appendix discusses "creative financing" which is generally defined as financing at a rate below market. This includes assumption financing, wraparound loans, buydowns, and below-market seller financing. You will learn of the value of financing to the borrower when market conditions and mortgage contract terms cause a transfer of wealth from the lender to the borrower. You should also better understand the implications that this type financing have on appraisals and the setting of market values. The problems for this chapter include assumption financing, buydown financing, and wraparound financing.

4A.1. **Cash Equivalence.** Calculate the Cash Equivalent values of the following mortgages

Mortgage	A	B	C	D
Original Term (MONTHS)	360	360	360	360
Contract rate	8%	7%	6%	7.5%
Remaining term	300	240	120	280
current market rate	12%	12%	11%	11%
Original balance	$100,000	$75,000	$80,000	$120,000

4A.2**. Assumption Financing**. Suppose that you want to buy a house with a purchase price of $120,000. The house has an assumable loan at 7.50% with 25 years remaining on its original 30 year term. The loan had an original balance of $100,000. The current market rate on new 25 year mortgages is 9.50%.

A. What is the cash equivalence value of the assumable loan?
B. What is the financing premium as indicated by cash equivalence that you (the buyer) would be willing to pay for the assumable loan?
C. What is the effective cost of the financing if the premium in part B is paid?
D. What is the effective cost of the financing if a financing premium of $8,000 is paid?
E. What is the effective cost of the financing if no premium is paid?

4A.3. **Buydown Financing**. To buy your dream home you need a mortgage for $100,000. The current market rate on thirty year, fixed-rate mortgages is 10 percent. This produces a monthly payment of $877.57 which you cannot afford. The seller offers to buy down the interest rate to 7 percent, 8 percent, and 9 percent, respectively, for the first three years.

A. What is the monthly payment for each of the first three years?
B. What amount must the seller pay up front?
C. What is the effective cost of the loan if you hold it to maturity?
D. What is the effective cost of the loan if you hold it for only six years?

4A.4**. Wraparound Financing**. Suppose you want to purchase a property for $150,000 and you have $20,000 to put down as a down payment. The property has an existing mortgage that can be wrapped. This loan is a fixed-rate mortgage at 7 percent, monthly payments. This loan had an original balance or $110,000 and has 20 years remaining on its original 30 year term. The current market rate for a new fixed-rate loan is 10.50 percent for 20 years. The seller will give you a wrap loan for the purchase price minus the down payment at 8.75 percent, monthly payments.

A. What is the effective equity yield for the wrap lender if the wrap loan is written for a term equal to the remaining term of the existing mortgage and both are held to maturity?
B. Suppose that, in order to lower the payments on the wrap loan, the wrap loan is written for a term of 30 years. Now, what is the effective yield for the wrap lender?
C. Assume the conditions in part B except that a balloon payment is required on the wrap loan at the end of 20 years (when the original loan matures). What is the effective equity yield for the wrap lender?

D. Suppose the wrap loan is written for 12 years and that the payments on the original loan are increased to equalize this 12-year maturity. What is the effective equity yield for the wrap lender?
E. Suppose the conditions in part D except that both loans are repaid in full after six years. What is the effective equity yield for the wrap lender?

CHAPTER 5
FIXED RATE MORTGAGE MECHANICS

This chapter explains the mechanics of the fixed-rate mortgage.

5.1. A borrower obtains a fully amortized fixed-rate mortgage for $125,000 at 8 percent interest for 30 years.
A. What is the monthly payment? B. What is the balance of the loan at the end of one year?

5.2. A 30 year mortgage loan was made five years ago for $80,000 at 10 percent interest. After winning the lottery, the borrower would like to prepay the mortgage balance by $8,000.

A. Assuming the loan maturity is shortened, what is the new loan maturity? (The payment stays the same.)
B. Assuming he can reduce his monthly mortgage payments, what is the new mortgage payment? (The maturity stays the same.)

5.3. A lender offers a fixed-rate mortgage under the following terms: $90,000 at 7.25% for 30 years, monthly payments. The lender will charge two discount points to make the loan. What is the annual percentage rate (APR) on this loan?

5.4. A borrower wants to buy a property for $140,000 and want a 90 percent loan for $126,000. A loan is available for 30 years at 7.50 percent interest, monthly payments. However, loan origination fees and discount points amount to $3,200.

A. How much money does the lender actually disburse?
B. What is the annual percentage rate (APR) on the loan?
C. If Jane pays the loan off after five years, what is the effective interest rate?
D. Assume that the lender imposes a prepayment penalty of 3 percent of the outstanding balance if the loan is repaid within six years. If Jane prepays the loan after five years with the prepayment penalty, what is the effective interest rate?

5.5. A lender extends a mortgage with the following terms: $130,000 at 8 percent for 30 years, monthly payments. However, the lender wishes to earn an APR of 8.50 percent.

A. How many discount points must the lender charge? (No other financing costs are present.)
B. How many points must the lender charge if the loan is expected to be repaid after ten years if the effective cost is 8.50 percent?

5.6. A borrower wants a 30-year fixed-rate mortgage for $80,000 and can choose between two options. Loan A has a 7 percent contract rate with five discount points (no other financing costs are charged on either loan). Loan B has a 7.50 percent contract rate with 1 discount point. What is the APR for each loan? Which is the better option, assuming that the borrower wishes to minimize his percentage cost of financing?

5.7. Suppose that a lender offers a fixed-rate mortgage at 9.50% for 30 years, monthly payments. You wish to borrow $75,000.

A. What is your monthly payment?
B. What is the balance at the end of the first year?
C. How much interest do you pay in the first year?
D. What is the balance if you decide to pay off the loan at the end of the sixth year?
E. How much total interest have you paid over this period?

5.8. You have just borrowed $90,000 at 7.50% for 15 years on a fixed-rate mortgage.

A. What is your monthly payment?
B. How much of the month one payment is interest?
C. How much of the month one payment is repayment of principal?
D. How much of the payment in month 120 is interest?
E. How much of the payment in month 120 is repayment of principal?
F. How much of the payment in month 180 is interest?
G. How much of the payment in month 180 is repayment of principal?

5.9. You borrow $80,000 for 30 years on a fixed-rate mortgage with monthly payments of $761.86. What interest rate are you paying?

5.10. You borrow an amount of money at 8.50% interest for 20 years with monthly amortization. Your payment is $607.48. How much did you borrow?

5.11. You want to purchase a house that is priced at $90,000. You can get a fixed-rate mortgage for 80 percent of the home's appraised value at 9.50% for 30 years, monthly amortization. The bank's appraiser has a theory that the value of the house is 95% of the asking price and appraises it accordingly. What will be your monthly payment if you take the loan?

5.12. Local lenders are offering the following terms for 30 year, fixed-rate mortgages. If you seek to minimize the effective cost of your borrowing, which alternative do you prefer?

Note that there is no loan amount given. You can pick whatever loan amount you wish. You will see that the loan amount has no effect on the effective cost of the loan. The solution assumes borrowing $1.

Option	Contract Rate	Discount Points
A	10.625%	2.75
B	10.75%	2.25
C	11.00%	1.50

A. Assuming monthly payments and that the mortgage will be held for its entire life, what is the effective cost (APR) of each alternative?

B. Assuming monthly payments and a holding period of five years, what is the effective cost of each alternative?

C. What is the effective cost of each alternative if the holding period is five years and a 3% prepayment penalty is present for each loan?

5.13. A borrower takes a fully amortized, fixed-rate mortgage loan of $100,000 at 12 percent interest for 30 years, monthly payments.

A. What is the monthly payment to satisfy interest and principal?
B. What is the monthly payment if the mortgage is written for a 15-year term?

5.14. What is the monthly payment required to fully amortize a $75,000, 10 percent mortgage over 15 years? What is the outstanding balance after 5 years?

5.15. A borrower borrowed $80,000 on a 30-year mortgage seven years ago. The fixed interest rate was 10 percent. The mortgage contract allows her to make partial pay-downs. She intends to pay off $10,000. When will the loan mature? (Note: the partial pay-down does not change the amount of the monthly payment.)

5.16. A house is offered for sale for $120,000 with an 80 percent mortgage for 30 years with fixed-rate interest at 8 percent monthly payments. The lender will charge 3 points.

A. How much will the lender effectively pay out?
B. If the mortgage is held to maturity, what will be the effective cost to the borrower?
C. Assume that there is a 2 percent prepayment penalty and the loan is paid off at the end of 5 years. What is the effective interest cost?

5.17. Mr. Jones is offered a fixed-rate loan of $100,000, monthly payment mortgage for 30 years. He has a choice between paying 11 percent interest and 3 points charged at closing or paying 10 percent with 5 points at closing. He expects to pay the loan off at the end of 8 years. If he wants to minimize his effective cost, which loan would be the better choice?

5.18. Suppose you take a 10/20 interest-only fixed-rate mortgage for $125,000 at 6 percent, monthly payments.
A. What is the interest-only payment?
B. What is the payment for the last twenty years to fully amortize the loan?
C. What is the balance of the loan at the end of year 25?

5.19. Suppose a borrower takes a 7/23 balloon/reset fixed-rate mortgage for $120,000 at 5 percent with an option to reset the rate at 6.50 percent when it balloons.
A. What is the payment for the initial period?
B. What is the balance of the loan at the reset?
C. What is the payment after the reset?

5.20. Your local lender offers you a fixed-rate mortgage with the following terms: $165,000 at 6.25% for 30 years, monthly payments. The lender will charge you two discount points and the loan has a 4% prepayment penalty.
A. What is the annual percentage rate (APR) of the loan?
B. What is the effective cost of the loan if it is repaid at the end of year 6?
C. How many points are required to yield an APR of 7.125%?

5.21 Suppose you take a $145,000 thirty-year fixed-rate mortgage at 6.50%, two discount points, monthly payments. At the end of the first year you inherit $12,000 from your now-favorite uncle. You decide to apply this $12,000 to the principal balance of your loan.
A. How many monthly payments are remaining after the extra lump sum payment is made?
B. What is your net interest savings over the life of the loan by doing this, assuming the loan is held for its full term?

5.22 You took a fixed-rate mortgage three years ago for $95,000 at 6.00% for 30 years, two discount points. Since your income is now higher you plan to pay an extra $125 per month on top of your normal payment. How many of the new payments are required to repay the loan?

5.23 Suppose you have two options on a $100,000, 30-year, fixed-rate mortgage. Option one is a 6.00% contract rate with 1.50 points. Option two is 5.75% contract rate but you have forgotten how many discount points are charged.
A. Calculate the number of points on option two that would equalize the APRs of the two loans.
B. Calculate the number of points on option two that would equalize the effective costs of the two loans over a five-year holding period.

CHAPTER 6
ALTERNATIVE MORTGAGE INSTRUMENTS

Chapter 6 deals with alternative mortgage instruments to the fixed-rate, fixed-payment mortgage. The primary alternative mortgage is the adjustable-rate mortgage (ARM). This chapter will help you understand the technical aspects of ARMs and other variable rate loans. You should also understand the pricing of this type mortgage and the effect of interest rate movements. The chapter will also help you understand other alternative mortgages such as the shared appreciation mortgage, the price-level-adjusted mortgage, the reverse annuity mortgage, and graduated-payment loans. The mechanics of mortgage refinancing are also discussed. Chapter 6 problems include: adjustable-rate mortgage (ARM), graduated payment mortgage (GPM), price-level adjusted mortgage (PLAM), shared appreciation mortgage (SAM), reverse annuity mortgage (RAM), flexible loan insurance program (FLIP), and refinancing of owner-occupied property.

6.1. **ARM.** You are offered a one-year ARM with an initial contract rate of 5.00%. Your contract calls for all subsequent adjustments to come from this initial rate (the interest rate caps, etc.). The margin on the loan is 2.50 and the index values are expected to be: EOY1 = 5.5%, EOY2 = 7.00%, and EOY3 = 8.75%. The loan has 2/6 interest rate caps. What will be your contract rate each for the next four years?

6.2. **ARM.** Suppose that you are considering an ARM with the following characteristics:

Mortgage Amount = $120,000
Index = One-year TB Yield
Margin = 2.50
Annual Interest Rate Cap = 2%
Lifetime Interest Rate Cap = 5%
Discount Points = 2
Loan Maturity = 30 Years

A. If the TB yield is currently 6%, what is the monthly payment for the first year? (Assume that that loan is fully indexed at the outset.)
B. If the index moves to 7.5% at the end of the first year, what is the payment for year 2?
C. If the loan is completely repaid at the end of year 2, what is the effective yield?
D. Suppose that, beginning in year two, the index remains at 7.5%. What is the APR on the loan?

6.3. **ARM.** You have been analyzing different ARM alternatives. You anticipate owning the home for five years. The lender offers a $100,000, 30 year ARM with the following terms:

Initial interest rate = 6%
Index = one-year TB yield
Payments adjusted once per year
Margin = 2.50
Interest rate caps = 2/6
Discount points = 2

Based on forward rates computed from the yield curve on Treasury securities, the index is estimated to have the values: EOY1 = 5%, EOY2 = 7.5%, EOY3 = 8%, EOY4 = 9.25%.

A. Compute the payments for the five-year period.
B. Compute the effective cost for the five-year holding period.
C. Assuming that the index value remains at 9.25% from EOY4 through EOY30, what is the effective cost (APR) of the loan?

6.4. **ARM.** Suppose you are considering an ARM with the following characteristics:

Mortgage amount = $85,000	2/6 Interest rate caps
Margin = 2.50	Index = 1 year TB yield
Term = 30 years	One-year adjustable
Monthly Payments	Discount points = 2
Initial Teaser Rate = 6.50%	Prepayment penalty = 3% for the first three years
All subsequent adjustments are made from the Teaser Rate	

A. If the TB yield is 6.00% at the outset, what is the monthly payment for year one?
B. If the TB yield goes from 6.00% to 7.75% at the end of year one, what is the monthly payment for year two?
C. If the loan is repaid at the end of year two, what is the effective cost?
D. Suppose that, at the end of year two, the index value is 8.00% and will stay this amount for the remaining term of the loan. If the mortgage is repaid at the end of year fifteen, what is the effective cost of the loan?

6.5. **ARM AND FRM.** Compare the following mortgages and determine which has the lower effective cost:

	Fixed-Rate	Adjustable-Rate
Mortgage amount	$100,000	$100,000
Term	30 years	30 years
Discount Points	2.5	3.75
Initial Contract Rate	10.75%	8.75%
Margin	----	2.75
Interest rate caps	----	2/6
Index value EOY0	----	8.75%
Holding period	3 years	3 years

Assume that all the ARM rate adjustments come from the initial contract rate and the index has these values: EOY1 = 8.75%, EOY2 = 10%, EOY3 = 11.75%.

6.6. **PLAM.** Suppose you are considering a PLAM which has the following features:

Mortgage amount = $95,000
Mortgage term = 30 years
Current real rate = 5%
Expected inflation rates = EOY1 = 2%, EOY2 = 6%, EOY3 = -3%
Annual payment adjustments

A. What is the monthly payment for year one?
B. What is the monthly payment for year two?
C. What is the monthly payment for year three?
D. What is the effective cost if the loan is repaid at the end of year three?
E. Suppose that, beginning in year four, the inflation rate is zero. What is the APR of the loan if it is held for its entire life?

6.7. **GPM.** Suppose that you borrow $90,000 for thirty years at 10% on a graduated payment mortgage. The rate of graduation is 7.50% beginning each year after origination for five years.

A. What is the monthly payment each year?
B. Set up a schedule to show the amortization of the loan for the first six months.
C. What is the balance of the loan at the end of year one?
D. What is the maximum amount that the loan balance increases to?
E. What would be the monthly payment if this were a standard, fixed-rate mortgage?
F. If two discount points are charged, what is the effective cost of the loan (APR)?
G. If this were a standard, fixed-rate mortgage, what would be the effective cost (APR)?

6.8. **SAM.** A shared appreciation mortgage (SAM) is written for $120,000 at a contract rate of 7% for thirty years, monthly payments. The lender has claim to 50 percent of the price appreciation in the property over the first five years. The property is currently valued at $135,000 and is expected to increase in value at a rate of 6 percent per year.

A. What is the monthly payment?
B. What is the lender's share of price appreciation at the end of five years?
C. Assuming the property is sold at the end of year five, what is the yield to the lender?
D. What rate of price appreciation would give the lender a return of 11 percent over the five years?
E. If the property appreciates at 6 percent per year and is sold at the end of year three, what is the lenders' yield?

6.9. **REVERSE MORTGAGE.** Your grandfather takes a reverse mortgage on his house for $100,000 at 8 percent for five years, annual annuity payments. What payment will he receive? How much interest does he pay on the loan? What is the balance owed if he decides to repay the loan at the end of year 2?

6.10. **REFINANCING.** Suppose that a borrower makes a mortgage loan for $120,000 at 9.50% for 30 years, monthly payments. This loan has a prepayment penalty of 4% of the outstanding balance. After five years the market rate on new mortgages is 7.25%. Lenders are charging 6% financing costs on new loans. The borrower's opportunity cost is 7.25%.

A. If the borrower plans to hold the loan to maturity (whether refinancing takes place or not), should he refinance if he wants to refinance the payoff of the existing loan for the remaining term of the existing loan?
B. Should the borrower refinance if he wishes to refinance the payoff of the existing loan for 30 years? Assume the contract rate on the new loan remains at 7.25%.
C. Should the borrower refinance if he wishes to refinance the payoff of the existing loan for 15 years? Assume that the contract rate on the new loan remains at 7.25%.
D. Suppose the borrower wants to have no out-of-pocket expenses and he wants the term on the new loan to equal the remaining term of the existing loan. Should he refinance? Assume the contract rate on the new loan remains at 7.25%.
E. Suppose the borrower wants to have no out-of-pocket expenses and he wants to extend the term on the new loan to 30 years. Should he refinance? Assume the contract rate on the new loan remains at 7.25%.
F. Assume the conditions in part A except the borrower knows that the holding period of the financing will only be six years from the date of refinancing. Should he refinance? Assume the contract rate on the new loan remains at 7.25%.
G. Suppose the borrower wants to take equity of $40,000 out of the house and he wants no out-of-pocket expenses. Should he refinance for a 25-year term? Assume that the contract rate on the new loan remains at 7.25%.

6.11. **INTEREST-ONLY ARM.** Suppose that you take a 3/1 interest-only ARM for $140,000 with an initial contract rate of 4.50 percent and a contract rate in year 4 of 6.00%. The loan is monthly payments for 30 years.
A. What is your payment for the first three years?
B. What is payment in year 4?
C. What is the balance of the loan at the end of year 4?

6.12. **ARM.** Suppose you are considering an ARM with the following characteristics:

Mortgage Amount = $135,000 Interest Rate Caps = 2/6 Margin = 2.50
Index = 1-year TB Yield Discount Points = 1.50 Term = 30 years
Monthly Payments

A. If the TB yield is currently 4%, what is the monthly payment for the first year? (Assume that the loan is fully indexed at the outset).
B. If the index moves to 5.50% at the end of the first year, what is the payment for year 2?
C. Suppose that, beginning in year two, the index remains at 5.50% for the life of the loan. What is the APR on the loan?

6.13. **ARM**. You are considering an ARM with the following characteristics:

Mortgage Amount = $125,000 2/6 Interest Rate Caps Margin = 2.50
Index = 1-year TB Yield Term = 30 Years One-Year Adjustable
Monthly Payments Discount Points = 2.25 Teaser Rate = 5.50%
Prepayment Penalty = 3% of the outstanding balance for the first three years
All subsequent interest rate adjustments are made from the teaser rate

A. If the TB yield is 5.00% at the outset, what is the monthly payment for year one?
B. If the TB yield goes from 5.00% to 6.75% at the end of year one, what is the monthly payment for year two?
C. If the loan is repaid at the end of year two, what is the effective cost?

6.14. **INTEREST-ONLY ARM.** Suppose you take a 3/1 interest-only ARM for $110,000 with an interest-only contract of 5.00 percent and a following rate of 6.50%.

A. What is your payment for the interest-only period?
B. What is the subsequent payment to fully amortize the debt over its stated term?
C. How may discount points would the lender have to charge to earn an APR of 7.00 percent?

6.15. **PLAM.** Suppose you are considering a PLAM with the following characteristics:

Mortgage Amount = $130,000 Term = 30 Years
Current Real Rate = 5.50% Annual Payment Adjustments
Expected Inflation Rates: EOY1 = 3%, EOY2 = -4%, EOY3 = 2%

A. What is your monthly payment for year one?
B. What is your monthly payment for year two?
C. What is the effective cost if the loan is repaid at the end of year two?

6.16. **REFINANCING.** Suppose that five years ago you took a mortgage loan for $180,000 at 7.750% for 30 years, monthly payments. This loan has a prepayment penalty of 5% of the outstanding balance for the first six years of life. The market rate on new mortgages now is 6.00%. Lenders are charging 6% financing costs on new loans. Your opportunity cost is 6.00%.

A. If you plan to hold the loan to maturity (whether refinancing takes place or not), should you refinance if
you want to refinance the payoff of the existing loan for the remaining term of the existing loan?
B. Should you refinance if you wish to refinance the payoff of the existing loan for 15 years? Assume that
the contract rate on the new loan remains at 6.00% and the loan is held to maturity.
C. Suppose you want to have no out-of-pocket expenses and you want to extend the term on the new loan to 30 years. Should you refinance? Assume the contract rate on the new loan remains at 6.00% and the loan is held to maturity.
D. Assume the conditions in part A except that you know that the holding period of the financing will only be six years from the date of refinancing. Should you refinance? Assume the contract rate on the new loan remains at 6.00%.
E. Suppose you want to take $30,000 of equity out of the house and you want no out-of-pocket expenses. Should you refinance for a 20-year term? Assume that the contract rate on the new loan remains at 6.00% and the loan is held to maturity.

CHAPTER 7
FEDERAL REGULATION OF FINANCIAL INSTITUTIONS RELATED TO THE MORTGAGE MARKET

The federal government has vast regulatory oversight of the financial markets. The United States Congress has, over a long period of time, passed laws that establish regulatory agencies that have oversight over financial markets. Occasionally, the U.S. Congress will change the regulatory structure by adding or removing regulatory agencies and laws and restructuring the regulatory environment. Although the regulatory environment encompasses the entire financial market this chapter will focus on those agencies and laws that have an impact on mortgage lending.

There are four components of financial regulation. One is safety and soundness. The safety and soundness of financial institutions has as its secondary goal a more stable and less risky economy. A second component is deposit insurance. An example would be the establishment of the Federal Deposit Insurance Corporation to insure the deposits at commercial banks (and now, savings and loan associations and mutual savings banks). The main purpose of deposit insurance is to prevent depositors from taking out their funds en mass or as in a "run on the bank". A widespread "run on the bank" could lead to economic recession or "panic" as money is removed from the economic system in large amounts. A third component is assurance of adequate capital for financial institutions. Capital, or equity, is basically, the net worth of a financial institution. Net worth, of course, is the value of the institution's assets less its liabilities. Adequate capital will help ensure that a particular or even a large number of institutions will not "fail". Lack of adequate capital will expose a financial institution to failure if, for example, its assets are risky (and therefore subject to a decline in market value) or its liabilities ate extremely short term (and subject to higher interest payments to depositors when the liabilities mature). The fourth component is a more recent one and is related to systematic risk. Briefly systematic risk refers to those risks that will affect all financial institutions simultaneously. The risk is widespread in the economy and exposes all financial institutions, not just single institutions, to risk. A good example of systematic risk is the real estate "bubble" that "burst" in 2006-2007 and caused widespread financial problems to numerous financial institutions. Non-systematic risk, on the other hand, affects only a single financial institution. An example would be the risk posed by incompetent management of the institution.

CHAPTER 8
FEDERAL HOUSING POLICIES: PART ONE

This chapter, along with Chapter 9, discusses federal policies toward the purchase and financing of residential properties. Four main categories are presented: (1) affordability programs to lower the cost of purchasing and/or financing residential properties, (2) programs to make the residential property market operate more efficiently, (3) policies to promote competition among the suppliers of residential properties, and (4) programs to make housing available to all citizens regardless of race, gender, national origin, handicap, etc. Disclosure laws pertaining to mortgage borrowing are discussed along with disclosure requirements such as annual percentage rate (APR). The problems for this chapter calculate the annual percentage rate (APR) for various loans.

8.1. **APR for Fixed-Rate Mortgage**. Suppose that you borrow $125,000 at 7.5 percent for 30 years, monthly payments. The lender charges you two discount points and one point origination fee. What is the annual percentage rate (APR) of the loan?

8.2. **APR on ARM**. You take an ARM for $90,000 for 30 years. The contract rate on the loan will be 5% for year one, 6.5% for year two, and 7.5% for years 3-30. If you pay two discount points, what is the APR of this loan?

8.3. **APR on Buydown Financing**. You borrow $140,000 at 8 percent for 30 years, monthly payments. The seller will buy your contract rate down to 6% the first and 7% the second year. The lender charges 1.50 discount points and 1 point origination fee. What is the APR of this loan?

CHAPTER 9
FEDERAL HOUSING POLICIES: PART II

This chapter completes the discussion of federal housing policies. The primary focus is on policies designed to protect the rights of minorities and the disadvantaged in the purchasing and financing of housing. These are mainly the Fair Housing Act and the Equal Credit Opportunity Act.

The Fair Housing Act (the Civil Rights Act of 1968) prohibits discrimination on the basis of race, color, religion, or national origin in the sale or rental of residential properties. Gender was added as a protected class in 1974. Families and the handicapped came under protection in 1988.

The Equal Credit Opportunity Act in 1974 prohibits discrimination in credit markets. The act prevents lenders from requesting information such as the number of children, alimony or child support, and marital status.

The Home Mortgage Disclosure Act of 1975 requires financial institutions of a certain size to compile a report on the distribution of its loans by census tract. The Community Reinvestment Act of 1978 requires all federally regulated financial institutions to publicize their lending activities within their community.

The Fair Housing Amendments Act of 1988 prohibits discrimination in renting based on age, number, and gender of children in the family.

CHAPTER 10
CONTROLLING DEFAULT RISK THROUGH BORROWER QUALIFICATION, LOAN UNDERWRITING, AND CONTRACTUAL RELATIONSHIPS

CHAPTER 11
LOAN ORIGINATION, PROCESSING, AND CLOSING

CHAPTER 12
MORTGAGE DEFAULT INSURANCE, FORECLOSURE, AND TITLE INSURANCE

Chapters 10, 11, and 12 discuss the mortgage borrowing process along with other topics such as government-sponsored methods of financing, private mortgage insurance, and mortgage default. Chapter 10 presents the various methods that lenders use to ensure against default risk. Within the context of borrower qualification, the guidelines for FHA and VA loans reviewed. Chapter 11 provides loan processing. You will understand the steps involved in processing a loan application and closing a mortgage. This chapter also discusses property appraisal and how it fits into the borrowing process. Chapter 12 presents three topics: mortgage default insurance (private, FHA, VA), the process of foreclosure, and title insurance. There are no problems from these chapters. However, the following outlines provide information on HUD programs, FHA loans, VA loans and private mortgage insurance (PMI).

HUD PROGRAMS

The Department of Housing and Urban Development provides various programs to assist homeowners. Examples of HUD-insured mortgage financing include (1) Energy Efficient Loans, (b) loans for rehabs, (c) loans for condominiums, (d) loans for disaster victims, and (e) reverse mortgages.

FHA LOANS

I. Components of Mortgage Credit Analysis

A. Determine credit risk to minimize foreclosure.

B. Cannot discriminate.

C. Lack of credit history is not basis for rejection.

D. Mortgage credit analysis includes:
   1. Determining the maximum loan amount.
   2. Estimating settlement requirements.
   3. Analyzing credit history.
   4. Calculating effective income.
   5. Estimating monthly housing expense.
   6. Assessing ability to repay mortgage and other liabilities in a timely fashion.

E. Borrower is rejected if

   1. Borrower cannot support payments.
   2. Bad credit history.
   3. Borrower is debarred from participating.
   4. Borrower is currently delinquent.
   5. A foreclosure does not automatically result in rejection but must be considered.

F. The borrower must pay a mortgage insurance premium (MIP). There is an up-front component and an annual premium (paid monthly) component of MIP. Starting January 1, 2001 the up-front premium is

1.50 percent and the annual premium is 0.5 percent of the outstanding balance. Annual premiums are nonrefundable. The up-front premium can be financed. The borrower can suspend the annual premium payment when the loan-to-value ratio reaches 78 percent of the original purchase price. Previously the up-front premium was 2.25% and the annual premium is 0.5% of the outstanding balance.

G. Principal Residence Requirement._Citizenship of U.S. is not required to borrow. The property must be the borrower's principal residence or his/her principal residence must be located in the U.S.

II. Amount and Amortization of Loan

A. Limitations
   1. The term cannot exceed 30 years.
   2. The current high-cost limit for a one-family residence is $290,319. FHA sets higher limits for areas with high housing costs. FHA will finance up to a four-family property as long as one unit is owner-occupied. FHA sets the limit at 95% of the median house price for a give SMSA or county.
   3. For non-high-cost areas, the limit is 48 percent of the FHLMC conforming loan limits. These limits are: (a) $271,050 for one unit, (b) $347,000 for a two-unit structure, (c) $419,400 for a three-unit structure, and (d) $521,250 for a four-unit structure.

B. HUD does not establish the maximum or minimum contract interest rate and does not regulate the amount of discount points paid by the borrower except for the Section 235 program (low income). Effective rate cannot vary more than 2% of local average.

C. Loan-to-value ratios
   1. If the application is accompanied by a Certificate of veteran status, 100% can be financed except for a $200 minimum. Discharge from the Armed Forces is required.

   2. FHA uses the following calculations to determine the loan amount:
      a. For a house with a price of $50,000 or less, 98.75% of the appraised value (closing costs excluded) or the sale price (excluding closing costs) whichever is less.
      b. For properties with appraised values (excluding closing costs) of greater than $50,000, 97.75% of lesser of the appraised value or the sale price (both excluding closing costs).

   3. When borrower and seller have family relationship such as parent and child and property is the borrower’s principal residence, the maximum mortgage amount is limited to 85% of appraised value.

   4. Secondary residences are limited to 85% of appraised value. Borrower can have only one.

   5. Borrower typically cannot have more than one high ratio loan at any one time. If the house goes from principal residence to rental, LTV ratio must be reduced to 75% based on current appraisal.

D. Acquisition Costs
   1. Only closing costs common to the area and normally paid by the buyer can be included in the calculation. Effective rate cannot vary more than 2% of local average.
   2. Prepaid items paid by the seller are deducted from the total cost of purchasing.
   3. Discount points are not included.
   4. Closing costs for the borrower include origination fee, discount points, prepaid interest, mortgage insurance premium, hazard insurance premium, title insurance, documentary stamps on deed, survey, deed preparation fee, recording fees, inspection, appraisal, credit report fee.

E. Payment-to-income ratios
   1. PITI cannot exceed 29% gross income.

2. Total expenses cannot exceed 41% of gross income.

F. No second mortgages are allowed.

G. Mortgage Assumptions
   1. Buyer must qualify over the entire life of the loan since December 1989.
   2. FHA allowed non-qualifying assumptions prior to December 1986.
   3. Between 1986 and 1989, FHA allowed non-qualifying assumptions after one year.
   4. Assumptor must be owner-occupant.

H. Lenders can now choose appraiser.

I. FHA can be refinanced up to 85% of appraised value plus closing costs. A borrower can refinance out of a graduated payment mortgage but not into one. Since the 1980s the FHA has permitted "streamline" refinances on insured mortgages. The streamline refers to the amount of documentation and underwriting required and does not mean a no-cost loan for the borrower. The requirements for streamlining are: (1) the mortgage to be refinanced must be FHA insured and must be current (not delinquent), (2) refinancing must result in a lower monthly principal and interest payment, and (3) no cash may be taken out in the refinancing. Some lenders offer "no-cost" refinances (borrower has no out-of-pocket expenses) by charging a higher interest rate. Other lenders may include the closing costs into the new mortgage amount. An appraisal determines whether there is sufficient equity in the property to permit this.

J. FHA has an adjustable-rate mortgage program. FHA ARMs are one-year adjustable, follow the Treasury bill yield index, have 1/5 caps, and a margin of 2.00.

K. FHA will finance manufactured homes.

L. FHA will also do reverse mortgages. Borrower must be at least 62 years old, own the property, occupy as principal residence, and participate in a consumer information session.

VA MORTGAGES

I. VA home loan program started in 1944 in the GI Bill of Rights. VA went to Cabinet rank in 1989.

II. Section 501 of Bill of Rights provides mortgage-lending guarantees for primary residences.

III. VA guarantees only part of the loan.

IV. Basic characteristics: owner-occupied, no down payment, limits on closing costs paid by the buyer, no prepayment penalty.

V. A funding fee is charged and can be financed into the mortgage. Fee varies with the down payment:

> 95% to 100% loan - 2.15%
> 90% to 95% loan - 1.50%
90% or less loan - 1.25%

Fee is higher for National Guardsmen and Reservists and is waived for veterans with service-related disability.

VI. Veteran must have <u>eligibility and entitlement</u>.

VII. Eligibility - must have served active duty the minimum time. 90 days for "hot" wars up to six years for reserves and National Guard.

VIII. Entitlement - amount of money VA will guarantee.
Basic Entitlement is $36,000
Veteran may have partial entitlement.
Entitlement travels with mortgage.

IX. Most VA loans are sold to Ginnie Mae that requires at least 25% guaranty.

X. Entitlement is restored by payoff of mortgage or substitution of another entitlement.

XI. Previously had non-qualifying assumptions. Rules changed in 1988. Now approval by lender is required.

XII. Release of liability is different from entitlement restoration. Example: mortgage could be assumed by non-veteran who qualifies.

XIII. In foreclosure, VA typically took title to property. Now it follows HUD/FHA procedure: get an appraisal then decide to take title or pay guaranty and not bid at the foreclosure sale.

XIV. Since 1992, market interest rates and discount points can be charged. Historically, the contract rate was fixed and no points (except funding fee) could be paid by borrower. Points cannot be added to loan amount.

XV. VA offers fixed-rate loans, graduated payment loans, and growing equity loans. Starting in 1992, VA has an ARM program. The terms are: 1/5 caps, TB yield index, margin of 2.00.

XVI. VA will also finance manufactured homes, home repair and improvement, and refinancing.

PRIVATE MORTGAGE INSURANCE (PMI)

I. Insurance provided by private insurers on conventional mortgages.

II. PMI insures the top portion of the loan where the down payment is less than 20%.

$$\text{Coverage Ratio} = \frac{\text{Loan Balance} - (0.75 \times \text{Value})}{\text{Loan Balance}}$$

III. In foreclosure, insurer can pay claim and take title to the property or it can simply pay the losses.

IV. PMI acts a co-insurance.

V. Typical Provisions:

A. Lender is required to follow the quickest foreclosure process.
B. Deed-in-lieu-of-foreclosure is encouraged to minimize legal expenses.
C. Attorney fees are limited to 3% of total claim.
D. Lender must make necessary repairs to restore property to its full value.

VI. The Homeowners Protection Act of 1998 requires lenders to cancel PMI automatically when the loan-to-value ratio is 78 percent of the original property value. Borrowers after July 29, 1999 can request cancellation at 80 percent loan-to-value ratio of the original property value. Automatic cancellation does not apply to loans originating before July 29, 1999 but the homeowners have to be notified of the right to do so.

VII. High-risk loans may not be canceled until the mid-way point of the loan (fifteen years on a thirty-year loan).

VIII. Other requirements for cancellation include: (1) no payments more than 30 days late in the last twelve months or no payments more than 60 days late in last twenty-four months and (2) property value has not declined from the original value.

IX. Some borrowers are using “piggyback” loans called 80-10-10s to avoid PMI. This is an 80 percent first mortgage, 10 percent second mortgage, and a 10 percent down payment.

FORECLOSURE

Foreclosure is the legal process by which a collateralized property is sold to repay the debt. The two methods of foreclosure are judicial foreclosure and power-of-sale. In a judicial foreclosure, the lender must proceed through the courts for permission to sell the property. In a power-of-sale state, a trustee is empowered to sell the property. The trustee must ensure that the rights of both parties are protected. Every state gives the defaulted borrower the right of redemption, which is the right to re-claim the property any time between default and the foreclosure sale. A few states allow the statutory right of redemption which allows the defaulted borrower to re-claim the property after the foreclosure sale. A deficiency judgment requires the defaulted borrower to pay any negative balance between the selling price of the property and the amount owed on the mortgage.

TITLE INSURANCE

Title insurance insures “good” title. This is title that is unencumbered and free from restrictions by private parties. Title insurance is issued by a title company after a title search.

CHAPTER 13
THE SECONDARY MORTGAGE MARKET

This chapter provides a description of the secondary mortgage market and the institutional participants. With this chapter, you should better understand what a secondary market is and how it differs from a primary market. You should also understand why the secondary mortgage market exists, why it creates value, and how its services are priced by the market.

The secondary mortgage is the market where existing mortgages are sold and securitized. In the secondary market mortgages are used as collateral to issue mortgage-backed securities. This provides a valuable flow of funds back to the primary mortgage market. This increases the funds available for mortgage lending. This reduces the primary market lender's reliance on deposits for funds available to lend.

There are various types of mortgage-backed securities: mortgage pass-throughs, mortgage-backed bonds, mortgage pay-through bonds, and collateralized mortgage obligations. The mortgage pass-through is the simplest form of security with no re-arranging of the cash flows. Other securities do re-arrange the cash flows since these bonds usually pay semi-annual interest and mortgage payments are collected monthly.

The major players in the secondary mortgage market are Fannie Mae, Freddie Mac, and Ginnie Mae. Fannie Mae was formed in 1968 to buy FHA loans. As of 1970, Fannie Mae is allowed to buy all three types of loans: FHA, VA, and conventional. Ginnie Mae was created in 1968 and does not buy mortgages, per se. Instead, it supports the FHA and VA loan market by guaranteeing pass-through securities. Freddie Mac was created in 1970 to provide a secondary market for conventional mortgages. Today Freddie Mac buys all three types of loans.

CHAPTER 14
MORTGAGE DERIVATIVE SECURITIES AND STRUCTURED FINANCE

This chapter presents the valuation of mortgage-related securities. With this chapter, you should better understand the structure of mortgage passthroughs, mortgage-backed bonds, and collateralized mortgage obligations. Prepayment of mortgages, the restructuring of cash flows on mortgage pools, and other factors that affect the value of secondary mortgage market securities are described. The problems for this chapter include mortgage passthroughs and mortgage-backed bonds.

14.1. Assume the following information about a pool of mortgages. Amount: $106,000,000, contract rate: 11%, maturity: 30 years. First year PSA= 100%, i.e. = .013. Provide the following data:

A. First year scheduled principal and interest payments separately.
B. The expected prepayment amount in the first year.
C. The first year end of year balance.
D. The total amount available for distribution to MBS security holders.

14.2. In the above problem assume that the first year PSA rate is 200% that is .026. Provide the following data:

A. The expected prepayment amount in the first year.
B. The first year end balance.
C. The total amount available for distribution to the security holders.

14.3. Continuing with this same problem, keep the PSA rate at 200% and change the contract rate to nine percent.
Now calculate for the first five years: the pool balance, scheduled P&I payments, prepayments and the amount owed to the security holders.

14.4. A GNMA 10 has a projected PSA of 200% at the current market rate of 9 percent. It is priced at 104. If the market interest falls to 8 percent the PSA rises to 225 and the price goes to 108. What is the effective or implied duration of this GNMA?

14.5. What would be the effective or implied duration in this example if the price of the GNMA rose to 109?

14.6. What is the value of a nine percent coupon (corporate) bond that promises to pay annual interest for thirty years if the market yield for this type of bond is 12 percent annually? The bond will return its face value of $1,000.

14.7. What is the value of the above bond if the market yield falls to 11 percent?

14.8. What is the duration of this bond?

14.9. **Mortgage Passthrough**. Assume the following pool of mortgages: $110,000,000 @ 9% for 30 years. For simplicity, assume annual mortgage payments.

<u>Part 1</u>
Construct the cash flows assuming no periodic prepayment. For brevity, assume that the mortgage pool is repaid in full at the end of year five.

<u>Part 2</u>
Construct the cash flows assuming 100% PSA. Again, assume that the mortgage pool is repaid in full at the end of year five.

Note: 100% PSA for each year is:

Year	Rate
1	1.3%
2	3.7%
3	5.75%
4	6.0%
5	6.0%

14.10. **Mortgage-Backed Bond.** Assume a mortgage pool of $108,000,000 at an 8% contract rate and a maturity of 30 years. Assume that the mortgage payments are annual payments. The pool will have no default. The mortgage pool will be used to collateralize a bond issue of $100,000,000 with a 7% coupon rate and a maturity of 20 years. The residual fund earns 6% annual return.

A. Assuming no periodic prepayment, construct the cash flows for the mortgage pool and the bond issue and calculate the net present value (at 8%) for the residual. For brevity assume that the mortgage balance and the bond balance are repaid in full at the end of year five.

B. Assuming a 100% PSA, construct the cash flows for the mortgage pool and the bond issue and calculate the net present value (at 8%) for the residual. Again, assume that the bonds and mortgage pool are repaid in full at the end of year five.

Note: 100% PSA for each year is:

Year	PSA
1	1.3%
2	3.7%
3	5.75%
4	6%
5	6%

CHAPTER 15
VALUE, LEVERAGE, AND CAPITAL STRUCTURE

CHAPTER 16
FEDERAL TAXATION AND REAL ESTATE FINANCE

These chapters present the cash flow analysis for income-producing real estate. Cash flows from operations and from resale are developed through to an after-tax basis; then, an analysis of the cash flows is presented. Chapter 15 begins the discussion by addressing the age-old question of the relationship between financing and property value. The concept of financial leverage is illustrated along with a discussion of agency theory. Chapter 16 summarizes tax legislation affecting income-producing real estate. There are several ways in which tax laws affect real estate: (a) the classification of income, (b) changes in marginal tax rates, (c) the treatment of capital gains, and (d) the treatment of tax shelter write-offs such as depreciation. The problems for these chapters show the cash flow analysis of income-producing real estate.

15/16.1. **Cash Flow Analysis**. A real estate investor is considering the purchase of a mini-warehouse. The investor knows the following information:

The purchase price is $800,000 with acquisition costs of $25,000.

There are 42,000 leasable square feet.

The rents are expected to be $6.50 per square foot per year and are expected to increase 5 percent per year.

The vacancy rate is expected to be 4 percent per year of gross rent.

Fees on late payments generate other income of $0.24 per square foot per year and will increase 5 percent per year.

Operating expenses are estimated at 45 percent of the effective gross income.

Eighty percent of the purchase price can be borrowed with a 20-year monthly payment mortgage at an interest rate of 12 percent plus 2 percent financing costs. Prepayment penalties are four percent of the outstanding balance.

The value of the investment is expected to increase 5 percent per year. Selling expenses will be 8 percent. The holding period is 5 years.

Of the total cost, 85 percent is depreciable.

The investor can be considered an active participant in the project and is in a 28 percent tax bracket. She requires an after-tax equity yield of 12 percent. Depreciation recovery is taxed at 25% and long-term capital gains are taxed at 15%.

A. Compute the ATCFs and the ATER for the holding period.

B. Calculate the NPV and IRR for the after-tax cash flows.

15/16.2. **Cash Flow Analysis**. A real estate investor is considering the purchase of an office building. The investor has the following information:

The purchase price is $900,000 with acquisition costs of $30,000.

The project is a two-story office building containing a total of 30,000 leasable square feet.

The rents are expected to be $10.75 per square foot per year and are expected to increase 5 percent per year.

The vacancy rate is expected to be 4 percent of gross rents per year.

Operating expenses are estimated at 42 percent of the effective gross income.

Eighty percent of the purchase price can be borrowed with a 20 year monthly payment mortgage at an interest rate of 10 percent plus 2 percent financing costs. There is a prepayment penalty of 4 percent of the outstanding balance for the first eight years of the mortgage's life.

The value of the investment is expected to increase 4 percent per year. Selling expenses will be 8 percent. The holding period is 5 years.

Of the total cost, 80 percent is depreciable.

The investor can be considered an active participant in the project and is in a 28 percent marginal tax bracket. The investor requires a 15 percent equity return. Depreciation recovery is taxed at 25% and long-term capital gains are taxed at 15%.

A. Compute the ATCFs and the ATER for the holding period.

B. Calculate the NPV and IRR. Is the project acceptable?

15/16.3. **Cash Flow Analysis**. A real estate investor is considering the purchase of an apartment triplex. The following information is known:

The purchase price is $115,000.

Each unit is expected to rent for $425 per month. Rents are expected to increase at a rate of 6 percent per year.

Vacancy and bad debt losses are estimated to be 3 months rental for one unit per year.

Operating expenses are expected to be 32 percent of EGI in year one with this dollar amount increasing at a rate of 8 percent per year.

Of the total cost, 85 percent is depreciable.

The project can be financed with a first mortgage of $85,000 at an interest rate of 11 percent for 20 years, monthly payments. Costs of placing the mortgage against the property (points, etc.) are estimated at 4 percent of the amount borrowed. There is a prepayment penalty of 6 percent of the outstanding balance if the loan is paid off within the first eight years of the mortgage life.

There is a second, interest-only mortgage of $10,000 with interest paid annually at 12 percent. The principal balance will be paid in full at the end of year three. An interest-only mortgage is one where the payment is interest and there is no amortization of the debt.

The investor expects to sell the investment for $142,000 at the end of 5 years with selling expenses equal to 8 percent of the selling price.

The investor is considered an active participant in the project and is in a 28 percent marginal tax bracket. The investor's required after-tax equity yield is 12 percent. Depreciation recovery is taxed at 25% and long-term capital gains are taxed at 15%.

A. Calculate the after-tax cash flows and then the NPV and IRR for the project.

B. Suppose that, instead of a fixed-rate mortgage, an ARM with the following features is used to finance the purchase:

Loan Amount	$95,000
Contract Rate	8.5% in year one, 10.5% in year two, 12% in year three, 11.5% in years four and five
Maturity	20 years
Monthly Payments	
Financing Costs	3.50% of the loan amount
No Second Mortgage	

Recalculate the after-tax cash flows and then the NPV and IRR.

16/16.4. **Cash Flow Analysis.** A real estate investor is considering the purchase of a house for rental property. The following information is known:

The purchase price is $220,000.

The house is expected to rent for $2,600 per month. Rent is expected to increase at a rate of 5 percent per year.

Vacancy and bad debt losses are estimated to be one month's rental per year.

Other income is expected to be 2 percent of gross rent.

Operating expenses are expected to be 30 percent of EGI in year one with this dollar amount increasing at a rate of 6 percent per year.

Of the total cost, 80 percent is depreciable.

The project can be financed with a first mortgage of $165,000 at an interest rate of 8 percent for 20 years, monthly payments. Costs of placing the mortgage against the property (points, etc.) are estimated at 5 percent of the amount borrowed. There is a prepayment penalty of 4 percent of the outstanding balance if the loan is paid off within the first six years of the mortgage life.

There is a second, interest-only mortgage of $20,000 with interest paid annually at 10 percent. The principal balance will be paid in full at the end of year three.

The investor's holding period is three years.

The value of the investment is expected to increase at a rate of 6 percent per year. Selling expenses equal to 8 percent of the selling price.

The investor is considered an active participant in the project and is in a 28 percent marginal tax bracket. The investor's required after-tax equity yield is 12 percent. Depreciation recovery is taxed at 25% and long-term capital gains are taxed at 15%.

Calculate the after-tax cash flows and then the NPV and IRR for the project.

15/16.5. **Installment Sale.** Smith sells property to Jones for $150,000 in a two-year installment agreement with 40 percent down and the balance financed over a five-year period at 10 percent annual interest. The adjusted basis of the property (original purchase price + capital improvements + acquisition costs minus accumulated depreciation) is $80,000. The property has been depreciated on a straight-line basis and has accumulated depreciation of $45,500. Smith has selling expenses of $5,000 and is in a 28 percent marginal tax bracket. Depreciation recovery is taxed at 25% and long-term capital gains are taxed at 15%. Calculate the after-tax cash flow in the year of sale and the after-tax cash flows from the installment receipts. Assume annual debt service payments. (Note that, regardless of the debt amortization period, all payments will be paid to the seller over the installment period. In other words, there will be a balloon payment at the end of year two.)

15/16.6. **Installment Sale.** Tom sells property to Mike for $220,000 in a two-year installment agreement with 30 percent down payment. Mike will assume an existing mortgage from Tom with annual payments of $3,064.21 at 6.50 percent with 12 years left on the life. The balance owed (purchase price minus the down payment and mortgage assumption) will be financed over a six-year period at 8 percent annual interest. Tom's original purchase price was $175,000 with acquisition costs of $5,000 and no capital improvements. The property has been depreciated on a straight-line basis and has accumulated depreciation of $66,000. Tom has selling expenses of $6,000 and is in a 28 percent marginal tax bracket. His tax rates on depreciation recovery and long-term capital gains are 25 percent and 15 percent, respectively. Calculate the after-tax cash flow in the year of sale and the after-tax cash flows from the installment receipts for the seller. Assume annual debt service payments.

15/16.7. **Tax-Deferred Exchange.** Larry has owned a property for eight years which is now worth $300,000. The property has an adjusted basis of $160,000 and Larry owes a mortgage balance of $140,000. He is seeking to expand his investment into a larger "like" property that is owned by Bob. This property is valued at $450,000 with a mortgage outstanding of $225,000 and an adjusted basis of $300,000. If they exchange properties, Larry will have selling expenses of $10,000 and Bob will have selling expenses of $16,000.

A. If these parties exchange properties, calculate the boot for each.

B. What is each investor's realized gain?

C. What is each investor's recognized gain?

D. What is the resulting tax basis for each property after the exchange?

15/16.8. **Incremental Cost of Borrowing.** Suppose an investor buys a property for $600,000. Lender A offers to finance 65 percent of the price with a 20 year, monthly payment mortgage at a rate of 6 percent. Lender B offers to make a 20 year, monthly payment loan for 80 percent of the purchase price but the contract rate will be 6.75 percent. What is the incremental cost of the additional loan amount across the two loans?

15/16/9. **Partitioning the IRR.** Suppose you have a real estate investment with the following cash flows:

Year	Initial Eq. Outlay	CF from Operations	CF from Resale
0	$500,000		
1		$50,000	
2		$75,000	
3		$100,000	$600,000

Partition the IRR and determine the percentage that each component of cash flow contributes to the total return.

15/16.10. **Optimal Capital Structure.** Suppose you have a real estate project that costs $1,125,000 and your required equity yield is 15% (regardless of the amount of debt). The project has the following equity cash flows under different levels of debt financing. Show the total value of the project (value of debt plus equity) for each of the debt levels. What is the optimal capital structure?

0% Debt Financing:

AFTER-TAX EQUITY CASH FLOWS

Year

	0	1	2	3	4	5
ATCF	-1,161,000	171,864	180,415	189,103	198,224	207,524
ATER						1,189,745

75% Debt Financing:

AFTER-TAX EQUITY CASH FLOWS

Year

	0	1	2	3	4	5
ATCF	-342,563	102,204	110,340	118,574	127,199	114,326
ATER						441,279

CHAPTER 17
ANATOMY OF A COMMERCIAL REAL ESTATE PROJECT: BEGINNING TO END

CHAPTER 18
REAL ESTATE IN A PORTFOLIO CONTEXT

This chapter discusses portfolio return and risk and the role that real estate plays in investment portfolios. Portfolio construction and the benefits of asset diversification are discussed.

Real estate must be evaluated on a return-risk basis, the same as other assets. The risk in real estate is measured as the variation in its return. The value of real estate to a diversified portfolio lies in its correlation of return with other assets. This will determine the extent to which the risk in real estate investment is diversifiable.

To evaluate real estate returns, there are various sources of data. These include the Russell-NCREIF Property Index, the NAREIT Equity Index, National Association of REALTORS® (NAR) data, and other sources.

Traditionally, real estate investors have sought to diversify by various means including property type, geographical diversification, and economic diversification.

CHAPTER 19
SOURCES OF FUNDS FOR COMMERCIAL REAL ESTATE PROPERTIES

This chapter discusses the equity and debt sources of funds for developing and purchasing income-producing real estate. Entities such as real estate investment trusts (REITs), real estate limited partnerships (RELPs), and pension funds are also discussed.

Institutions invest in real estate by either lending funds directly or by purchasing debt obligations from another originator. For a given institution, the net investment in its end-of-year holding of debt securities will equal its originations plus its purchases less sales and less repayment of debt outstanding from the previous year. There are a number of sources of commercial debt, including life insurance companies, commercial banks, and savings institutions. Commercial mortgage-backed securities (CMBSs) are becoming an increasingly greater source of debt financing for commercial properties.

CHAPTER 20
ACQUISITION, DEVELOPMENT, AND CONSTRUCTION FINANCING

This chapter deals with acquisition, development, and construction financing. The role of the developer is quite different from the real estate investor. The developer requires funds for short-term development and not for long-term investment. A "take-out" loan (long-term loan to pay off the short-term loan) is generally required for short-term loan approval. The structure of ADC loans is discussed. The problem for this chapter is a construction loan problem.

18.1. **Construction Loan Problem.** Suppose that Ace Construction Co. wishes to build a 20 unit apartment complex on land it owns free and clear that is worth $150,000. Ace has applied for and received a "takeout" commitment from a local permanent lender in the amount of $475,000, pending completion of the project and proof of rental of 16 units at an agreed upon rent. An appraisal report indicates that the complex will have a value of $675,000 at completion.

The following costs are estimated by Ace:

1. Constructing the building ($22/sq. ft. for 16,000 sq. ft.)	$352,000
2. Fixtures	30,000
3. Landscaping	18,000
4. Water/Sewer	30,000
5. Closing Costs/Taxes	6,500
6. Loan Origination	14,250
7. Architect Fees	6,000
8. Inspections	2,000
Total	$458,750

Assume that a construction loan is made in the amount of $458,750 at 12 percent annual interest and that items 5, 6, and 7 are paid at closing. If the construction takes exactly 12 months to complete and the developer plans to draw the balance (what is left after closing) in equal monthly draws, complete a draw schedule for the builder and calculate the effective yield for the lender.

CHAPTER 21
PERMANENT FINANCING OF COMMERCIAL REAL ESTATE PROPERTIES

This chapter reviews long-term, permanent financing for income-producing real estate. The discussion focuses primarily on equity participation loans, lease versus buy decisions, and sale-leaseback arrangements. The problems for this chapter include equity participation loans, lease versus own, and sale-leasebacks.

21.1. **Equity Participation Loan.** A real estate investor is considering the purchase of an office building using an equity participation loan. The following information is known:

Project Cost:	
Land	$120,000
Building	$680,000
Total	$800,000
Project financing:	
Loan amount	$640,000
Interest rate	9%
Amortization	20 Years
Participation terms:	
Share of NOI	20% of NOI above $100,000
Share of price appreciation	12%
Initial equity investment	$172,800
Gross rent: Year 1	$252,000
Other income:	8% of gross rent each year
Vacancy rate	8%
Operating expense:	45% of EGI
Growth rates:	
Gross rent	5%
Resale price	5%
Selling Expenses	8%
Investor's marginal tax rate	28%
Holding period:	5 years

A. Calculate the after-tax cash flows from operations and the after-tax equity reversion.

B. Calculate the NPV and IRR for the borrower and the IRR for the lender.

21.2. **Lease Versus Own.** You are about to experience your life-long dream of owning a pizza parlor and you have found the perfect location. One decision you have to make is whether to lease or buy the property for the next five years. The owner is willing to rent you the property on a triple net lease (you pay property taxes, insurance, and maintenance). If you buy, you must incur a mortgage but you will have the depreciation and any price appreciation. If you lease, you have a fully tax-deductible lease payment, but you can't depreciate and you receive no price appreciation. The characteristics of the property are:

Purchase Price	$420,000
Land Value	$84,000
Building Value	$336,000
Selling Price EOY5	$560,000
Interest-Only Mortgage	$315,000 with 9.5% interest paid annually and principal balance paid at EOY5
Marginal Tax Rate	28%
Required Equity Yield	15%
Lease Payment If Leased	$31,000 per year, paid at the end of each year for 5 years

A. Which alternative do you prefer?

B. What lease payment would make you indifferent between the two alternatives?

C. Which alternative is better if you successfully negotiate to pay the breakeven lease payment but it is required to be an annuity due?

21.3. **Sale-Leaseback**. Assume you own the following non-residential property and are deciding whether to continue to own the property or execute a sale-leaseback. The following information is available:

Original cost:	$975,000
Book value:	$225,000
Market value:	$1,200,000
Debt balance:	$65,000
Rate	10%
Remaining term	10 years
Payment	$10,578
Investor's tax rate:	28%
Annual depreciation:	$25,000 with 5 years remaining
Yearly NOI	$226,000
Growth rates:	0%

If you do a sale-leaseback, the agreement calls for: $150,000 annual lease payments and you will sell the property for $1,200,000 and repurchase it for $1,300,000 after five years.

A. Calculate the after-tax cash flows from operations assuming the property is retained as is.

B. Calculate the after-tax cash flows assuming the property is sold and leased back.

C. Calculate the net present values for the two alternatives. Which is preferable?

21.4. **Equity Participation Loan.** Assume the following data for an equity participation loan:

Project Cost: Land, $300,000, Building, $2,500,000

Loan Amount: $1,800,000

Interest rate and period, 9.5%, 30 years, annual payments

Share of NOI: 18% over $250,000

Share of Appreciation: 15%

Year One Gross Rent: $525,000

Year one operating Expense: 16% of gross rent

Vacancy Rate: 5% annually

Growth Rates: rent, 8% operating expenses, 7%

Depreciation: $87,121 in year one, $90909 thereafter.

Investor's marginal Tax Rate: 28%

A. Calculate the before- and after-tax cash flows from operating the facility for a five year period

B. Calculate the cash flows to the lender for the first four years.

Note: Do not calculate any cash flows associated with a sale of the property.

CHAPTER 22
OWNERSHIP STRUCTURES FOR FINANCING AND HOLDING REAL ESTATE

This chapter discusses the various ownership forms for real estate ranging from sole proprietorship to partnerships to real estate investment trusts. The preferred type depends primarily on tax laws, accessibility to the capital markets, and issues of personal liability.

A sole proprietorship is ownership by one person. An advantage is no double income taxation but a sole proprietor typically does not have access to the large equity capital markets and there is no limited liability. The corporate form may be either a C corporation or an S corporation. While both provide limited liability, double taxation is present in the C Corporation and absent in the S corporation.

Partnerships are associations of at least two people and may be general or limited partnerships. In a general partnership, all partners are responsible for the obligations incurred by the partnership. A limited partnership must contain at least one general partner who is responsible for all the debts incurred whereas the limited partners are shielded.

Real estate investment trusts (REITs) are corporations formed to invest in real estate. REITs are designed to provide a way for smaller investors invest in real estate through a professionally managed company. Special rules for REITs allow them to avoid the double taxation that is typical with corporations. There are several types of REITs: equity, mortgage, and hybrid.

SOLUTIONS TO PROBLEM GUIDE PROBLEMS

In order to work the problems in the workbook, you should know the following time value of money equations and expressions.

FVF = Future Value of One Dollar Factor
PVF = Present Value of One Dollar Factor
i = interest rate
n = number of payments
FVAF = Future Value of an Annuity Factor
FVAFD = Future Value of an Annuity Due Factor
PVAF = Present Value of an Annuity Factor
PVAFD = Present Value of an Annuity Due Factor
SFF = Sinking Fund Factor
MC = Mortgage Constant Factor

Future Value of One Dollar (Lump Sum) Factor

$$FVF = (1 + i)^n$$

Present Value of One Dollar (Lump Sum) Factor

$$PVF = 1/ (1 + i)^n$$

Future Value of an Annuity Factor

$$FVAF = \frac{(1 + i)^n - 1)}{i}$$

Future Value of an Annuity Due Factor

$$FVAFD = [((1 + i)^n -1)/i)] (1 + i)$$

Present Value of an Annuity Factor

$$PVAF = \frac{(1 + i)^n - 1}{i (1 + i)^n}$$

Present Value of an Annuity Due Factor

$$PVAFD = [((1 + i)^n - 1)/(i (1 + i)^n)] (1 + i)$$

Sinking Fund Factor

$$SFF = i / ((1 + i)^n - 1)$$

Mortgage Constant Factor

$$MC = \frac{i (1 + i)^n}{(1 + i)^n - 1}$$

Mortgage Constant Factor/Monthly Payment

$$MC = \frac{(i/12) (1 + i/12)^{12n}}{(1 + i/12)^{12n} - 1}$$

SOLUTIONS TO TIME VALUE OF MONEY PROBLEMS

The financial calculator keystrokes to derive the solutions are shown. All cash outflows are negative (as when an investment is purchased). All cash inflows are positive (as when an investment is sold). Answers may be rounded. The calculator should be set to one payment per year and should not be in the "Begin" mode.

A.1. Old Bank: $FV_{EOY5} = 1{,}000\ (1.06)^5 = \$1{,}338.23$

New Bank: $FV_{EOY5} = 1{,}000\ (1 + .06/12)^{60} = \$1{,}348.85$

Difference: $10.62

<u>Calculator Keystrokes</u>

Old Bank:	New Bank:
-1,000 PV	-1,000 PV
6 I/YR	6/12 I/YR
5 N	60 N
Press **CPT FV**	Press **CPT FV**

A.2. Option One: $FV_{EOY1} = 10{,}000\ (1.08)^1 = \$10{,}800$ i = 8%

Option Two: $FV_{EOY1} = 10{,}000\ (1 + i/4)^4 = \$10{,}800$ i = 7.77%

Option Three: $FV_{EOY1} = 10{,}000\ (1 + i/12)^{12} = \$10{,}800$ i = 7.72%

<u>Calculator Keystrokes</u>

-10,000 PV	-10,000 PV	-10,000 PV
8 I/YR	10,800 FV	10,800 FV
1 N	4 N	12 N
CPT **FV**	**CPT I/YR** x 4	**CPT I/YR** x 12

A.3. $PV = 15{,}000/1.14 + 15{,}000/(1.14)^2 + 15{,}000/(1.14)^3 + 15{,}000/(1.14)^4$

$+ 165{,}000/(1.14)^5$

= $129,402

<u>Calculator Keystrokes</u>

15,000 PMT
150,000 FV
14 I/YR
5 N
CPT **PV**

A.4. $PV = \dfrac{80{,}000}{(1 + .15/4)^{20}} = \$38{,}311.39$

80,000 FV
20 N
15/4 = 3.75 I/YR

CPT **PV**

A.5. PV = $5{,}000/1.16 + 5{,}000/(1.16)^2 + \ldots + 5{,}000/(1.16)^{10}$

= $5{,}000\ (PVAF_{16,10})$

= $24,166.14

5,000 PMT
10 N
16 I/YR
CPT **PV**

A.6. A. $10{,}000 = 3{,}000\ (PVAF_{i,5})$ i = 15.24%

B. $10{,}000 = 3000\ (PVAFD_{i,5})$ i = 25.68%

A. -10,000 PV
3,000 PMT
5 N
CPT **I/YR**

B. Set calculator to "Begin" and repeat inputs

A.7. $PV = \dfrac{98{,}000}{(1.06)^8}$

98,000 FV
6 I/YR
8 N
CPT **PV**

A.8. A. $FV_{EOY10} = 800\ (1.08)^{10} = \$1{,}727.14$ per acre

B. $1{,}400 = 800\ (1 + i)^{10}$ i = 5.76%

A. -800 PV
10 N
8 I/YR
CPT **FV** per acre

B. -800 PV
10 N
1,400 FV
CPT **I/YR**

A.9. A. $(.70)\ 100{,}000\ (MC_{10,8}) = \$13{,}121.08$

B. Payment $= 70{,}000\ (MC_{10,8}) = 70{,}000\ \left[\dfrac{(.10/12)\ (1 + .10/12)^{96}}{(1 + .10/12)^{96} - 1}\right] = \$1{,}062.19$

A. -70,000 PV
10 I/YR
8 N
CPT **PMT**

B. -70,000 PV
10/12 I/YR
96 N
CPT PMT

A.10. $FV_{EOY5} = 120\ (FVAF_{5.75/12,\ 60}) = \$8318.83$ saved

.20x = 8,318.83

x = $41,594

120 PMT
5.75/12 I/YR
60 N
CPT **FV**
FV/.20 = $41,584

A.11. 100,000 = Payment ($FVAF_{6/12,\ 120}$)

Payment = $610.20

100,000 FV
6/12 I/YR
120 N
CPT **PMT**

A.12. PV = 25,000 ($PVF_{25,3}$) = $12,800 per acre

25,000 FV
25 I/YR
3 N
CPT **PV** per acre

A.13. PV of Annuity Due = 1,500 ($PVAFD_{10/12,\ 60}$) = $71,186

Set calculator to "Begin"
1,500 PMT
60 N
10/12 I/YR
CPT **PV**

A.14. PV = (1,800 - 1,000) ($PVAFD_{12/12,60}$) = $36,324

Set calculator to "Begin"

1,000 PMT	1,800 PMT
60 N	60 N
12/12 I/YR	12/12 I/YR
CPT **PV**	**CPT PV**

A.15. PV = $20{,}000/1.15 + 22{,}000/(1.15)^2 + 25{,}000/(1.15)^3 + 30{,}000/(1.15)^4$

$+ 32{,}000/(1.15)^5$

= $83,527

This problem is worked in the cash flow mode of the calculator.
0 CF0
20,000 CF1
22,000 CF2
25,000 CF3

30,000 CF4
32,000 CF5
15 I/YR
Shift Key NPV

A.16. A. $FV_{EOY5} = 5{,}000\ (1.08)^5 = \$7{,}347$

B. $FV_{EOY5} = 5{,}000\ (1 + .08/4)^{20} = \$7{,}430$

C. 7,430 - 7,347 = $82.73

D. Option A is 8%, Option B is 8.24% = $(1.02)^4 - 1$

A.	B.	C.	D.
-5,000 PV	-5,000 PV	-5,000 PV	-5,000 PV
8 I/YR	8/4 I/YR	5 N	20 N
5 N	20 N	7,347 FV	7,430
CPT FV	**CPT FV**	**CPT I/YR**	**CPT I/YR** x 4

A.17. $FV_{EOY1} = 50{,}000\ (1 + .06/12)^{12} = \$53{,}084$

$FV_{EOY1} = 50{,}000\ (1.07)^1 = \$53{,}500$

You prefer the 7% option

-5,0000 PV	-50,000 PV
12 N	1 N
6/12 I/YR	7 I/YR
FV	**FV**

A.18. A. PV = 30,000 ($PVF_{10,8}$) = $13,995 No

B. 30,000 = 15,000 ($FVF_{i,8}$) i = 9.051%

A.	B.
30,000 FV	-15,000 PV
8 N	30,000 FV
10 I/YR	8 N
CPT **PV**	**CPT I/YR**

A.19. PV = 80,000 ($PVF_{14,4}$) = $47,366

80,000 FV
4 N
14 I/YR
CPT **PV**

A.20. A. Future Value of an Annuity = 4,000 ($FVAF_{6/2,20}$) = $107,481

B. Future Value of an Annuity Due = 4,000 ($FVAFD_{6/2,20}$) = $110,706

A. 4,000 PMT B. Set calculator to “Begin” and repeat calculations

6/2 I/YR
20 N
CPT **FV**

A.21. Future Value of an Annuity = 2,000 ($FVAF_{10/12,72}$) = $196,223

2,000 PMT
10/12 I/YR
72 N
CPT **FV**

A.22. FV_{EOY10} = 1,000 $(1.06)^9$ + 2,000 $(1.06)^6$ + 500 $(1.06)^5$

= 1,689 + 2,837 + 669

= $5,195

-10,000 PV	-2,000 PV	-500 PV
9 N	6 I/YR	6 I/YR
6 I/YR	6 N	5 N
CPT **FV +**	**CPT FV**	**CPT FV**

A.23. PV = 5,000 ($PVAF_{14,10}$) = $26,080

5,000 PMT
10 N
14 I/YR
CPT **PV**

A.24. A. 50,000 = Payment ($FVAF_{8,10}$)

Payment = $3,451.47

B. 50,000 = Payment ($FVAF_{8/2,20}$)

Payment = $1,679.09

A.	50,000 FV	B.	50,000 FV
	10 N		20 N
	8 I/YR		4 I/YR
	CPT **PMT**		**CPT PMT**

A.25. FV = 100,000 ($FVF_{7,10}$) = $1,967,151

-1,000,000 PV
7 I/YR
10 N
CPT FV

A.26. 1,000,000 = 2,000,000 ($PVF_{i,10}$) = 7.177%

-1,000,000 PV
2,000,000 FV
10 N
CPT I/YR (note: if you get "no solution" you forgot to make either the PV or the FV negative and the other positive)

A.27. 100,000 = 30,000 ($PVAF_{i,10}$) + 2,000,000 ($PVF_{i,10}$) = 9.44%

-1,000,000 PV
2,000,000 FV
30,000 PMT
10 N
CPT I/YR

A.28. 100,000 = PMT ($PVAF_{12,5}$) + 1,500,000 ($PVF_{12,5}$) = $41,295

-1,000,000 PV
1,500,000 FV
5 N
12 I/YR
CPT **PMT**

A.29. PV = 40,000 ($PVAF_{12,5}$) = $144,191

-$40,000 PMT
5 N
12 I/YR
CPT PV

A.30. PV = 40,000 ($PVAF_{12,5}$) = $161,491

Set calculator to "Begin" mode and repeat calculations
-40,000 PMT
4 N
12 I/YR
CPT PV and add $40,000

A.31. PV = 40,000 ($PVF_{13,1}$) + 40,000 (1.04) ($PVF_{13,2}$) + 40,000 $(1.04)^2$ ($PVF_{13,3}$)

+ 40,000 $(1.04)^3$ ($PVF_{13,4}$) + 40,000 $(1.04)^4$ ($PVF_{13,5}$) = $150,955

0 CF0
40,000 CF1
41,600 CF2
43,264 CF3
44,995 CF4
46,794 CF5
13 I/YR
Shift Key NPV

A.32. FV = 50,000 ($PVAF_{8,10}$) = $782,274

$50,000 PMT
10 N
8 I/YR
CPT **FV**

A.33. Lease A. PV = 100,000 ($PVAF_{12,5}$) = $222,054

$55,000 PMT
5 N
12 I/YR
CPT PV

Lease B. PV = 45,000 ($PVAF_{12,5}$) + 35,000 = $216,681

$45,000 PMT
5 N
12 I/YR
CPT PV
add $35,000
accept lease B

A.34. PV = 100,000 ($PVAF_{9,9}$) = $599,525 599,525 = PMT ($FVAF_{9,10}$) = $39,461

$100,000 PMT
9 I/YR
9 N
CPT PV
answer is $599,525

$599,525 FV
9 I/YR
10 N
CPT PMT

A.35. 599,525 = PMT ($FVAF_{6,10}$) = $45,485

$599,525 FV
6 I/YR
10 N
CPT PMT

A.36. $2 = 1 (1.07)^n = 10.24\%$

-1 PV
2 FV
7 I/YR
CPT N

A.37. $2 = 1 (1 + i)^8 = 9.05\%$

-1 PV
2 FV
8 N
CPT I/YR

A.38. PMT = 100,000 ($MC_{9,30}$) = $9,733.64

-100,000 PV
30 N
9 I/YR
CPT PMT

A.39. $5,000 (1.08)^{10} = \$10,795$ 45,000 - 10795 = PMT ($FVAF_{8,10}$) = $2,361

-$5,000 PV
10 N
8 I/YR
CPT FV
answer $10,794.62
shortfall = $34,205.38

$34,205.38 FV
10 N
8 I/YR
CPT PMT

A.40. $FV = 10,000 (1.08)^5 = \$14,693$ $8,000 (1.09)^4 = \$11,293$

$6,693 (1.08)^4 = \$9,106$ 11,293 + 9,106 = $20,398.78

-10,000 PV
5 N
8 I/YR
CPT FV
answer $14,693.28

less $8,000
= $6,693.28

-$6,693.28 PV
4 N
8 I/YR
CPT FV
answer $9,106.13

-$8,000 PV
4 N
9 I/YR
CPT FV
answer $11,292.65

$11,292.65 + $9,106.13 = $20,398.78

SOLUTIONS TO CHAPTER TWO PROBLEMS

2.1. Assuming that the Fisher Equation is accurate, provide the missing data in the following table.

Scenario	Nominal Interest Rate	Real Interest Rate	Expected Inflation
A	10%	4%	**6%**
B	9%	**5%**	4%
C	**6%**	3%	3%
D	8%	**3%**	5%

2.2. Using the Equation of Exchange, fill in the missing values in the table.

Scenario	Money Supply	Velocity	Index of Price Level	Real Transactions	GNP
A	**1500B**	5	**10**	750B	7500B
B	**1600B**	4	8	800B	**6400B**
C	500B	**6**	10	**300B**	3000B
D	500B	3	5	300B	**1500B**
E	**600B**	5	6	**500B**	3000B

2.3. If the velocity of money remains constant in the Equation of Exchange, what will the rate of inflation be under the following conditions:

Current money supply = $2,000B
Velocity of circulation = 6
Real transactions = $3,000B
Price level index = 4
Next year's money supply = $2,400B
Next year's real transactions = $3,200B

Money Supply	Velocity	Price Level Index	Real Transactions
2000B	6	4	3000B
2400B	6	**4.5**	3200B

$$\frac{4.5 - 4}{4} = .125 \text{ or } 12.5\% \text{ rate of inflation}$$

2.4. Fill in the missing values in the following table.

Scenario	Nominal Rate	Risk Free Rate	Default Risk Premium	Callability Risk Premium	Maturity Risk Premium
A	10%	4%	**2%**	2%	2%
B	**11%**	3%	3.50%	1.50%	3%
C	9.50%	4%	**0.5%**	3%	2%
D	8%	**2.5%**	3%	1.50%	1%
E	12%	4%	3%	3%	**2%**

2.5. If the current rate on a one-year bond is 6% and the current rate on a two-year bond is 4%, what is the expected rate on a one-year bond one year from now?

$(1 + {}_1r_0)(1 + {}_1r_1) = (1 + {}_2r_0)^2$

$(1.6)(1 + {}_1r_1) = (1.04)^2$

$1 + {}_1r_1 = \frac{1.0816}{1.06}$

$1 + {}_1r_1 = 1.0204$

${}_1r_1 = 2.04\%$

2.6. If the current rate on a one-year bond is 8%, the current rate on a two-year bond is 12%, and the current rate on a three-year bond is 15%, what is the expected rate on a one-year bond at the end of year one? What is the expected rate on a one-year bond at the end of year two?

$(1.08)(1 + {}_1r_1) = (1.12)^2$

$1 + {}_1r_1 = 1.161$

${}_1r_1 = 16.1\%$

$(1.15)^3 = (1.12)^2 (1 + {}_1r_2)$

$1.521 = (1.254)(1 + {}_1r_2)$

${}_1r_2 = 21.3\%$

2.7. A. $(1.055)^2/(1.05) = 1.06 = 6\%$
B. $(1.06)^3/(1.055)^2 = 1.07 = 7\%$
C . $(1.063)^4/(1.06)^3 = 1.072 = 7.2\%$
D. $(1.064)^5/(1.063)^4 = 1.068 = 6.8\%$

2.8. $(1.045)^3/(1.05)^2 = 1.030166 = 3.02\%$

2.9. seven percent, that is, 7% = 2% + 5%

2.10. three percent, that is, 7% = 3% + 4%

2.11. two, that is, 4 x 500 B = 1,000 B x 2

2.12. 2.1153, that is, 4 x 550 B = 1040 x 2.11543

2.13. Inflation is 5.77 percent, that is, 2.1153/2 = 1.0577

SOLUTIONS TO CHAPTER FOUR APPENDIX PROBLEMS

4A.1.

Mortgage	A	B	C	D
payment	$733.76	$498.98	$479.64	$839.06
current balance	$95,069.26	$64,359.29	$43,202.87	$110,792.78
new payment	$1001.30	$708.65	$595.12	$1,101.15
payment savings	$267.53	$209.67	$115.48	$262.10
cash equivalent (PV)	$25,401.44	$19,042.50	$8,383.24	$26,370.87

4A.2.

A. Pmt = 100,000 ($MC_{7.5,30}$) = $699.21

Bal_{EOY5} = $94,617

Cash Eq. Value = 699.21 ($PVAF_{9.5/12,300}$) = $80,029

B. Financing Premium = 94,617 - 80.029 = $14,588

or alternatively,

Pmt_{NEW} = 94,617 ($MC_{9.5,25}$) = $826.66

Payment Savings = 826.66 - 699.21 = $127.46

Present Value of Payment Savings = 127.46 ($PVAF_{9.5/12,300}$) = $14,588

C. 94,617 - 14,588 = 699.21 ($PVAF_{i/12,300}$) i = 9.5%

D. 94,617 - 8,000 = 699.21 ($PVAF_{i/12,300}$) i = 8.53%

E. 94,617 - 0 = 699.21 ($PVAF_{i/12,300}$) i = 7.5%

4A.3.

A. Pmt_1 = 100,000 ($MC_{7,30}$) = $665.30
Pmt_2 = 100,000 ($MC_{8,30}$) = $733.76
Pmt_3 = 100,000 ($MC_{9,30}$) = $804.62

B. Present Value of Difference in Payments = (877.57 - 665.30) ($PVAF_{10/12,12}$)

+ (877.57 - 733.76) ($PVAF_{10/12,12}$) ($PVF_{10/12,12}$)

+ (877.57 - 804.62) ($PVAF_{10/12,12}$) ($PVF_{10/12,24}$)

= $4,575

C. 100,000 = 665.30 ($PVAF_{i/12,12}$) + 733.76 ($PVAF_{i/12,12}$) ($PVF_{i/12,12}$)

+ 804.62 ($PVAF_{i/12,12}$) ($PVF_{i/12,24}$) + 877.57 ($PVAF_{i/12,324}$) ($PVF_{i/12,36}$)

i = 9.47%

D. Bal_{EOY6} = $95,660

100,000 = 665.30 ($PVAF_{i/12,12}$) + 733.76 ($PVAF_{i/12,12}$) ($PVF_{i/12,12}$)

$+ \ 804.62 \ (PVAF_{i/12,12}) \ (PVF_{i/12,24}) \ + \ 877.57 \ (PVAF_{i/12,36})$

$+ \ 95{,}660 \ (PVF_{i/12,72})$

$i = 8.98\%$

4A.4.

A. $Pmt_{EX.\ LOAN} = 110{,}000 \ (MC_{7,30}) = \$731.83$

$Bal_{EOY10} = \$94{,}394$

$Pmt_{WRAP} = 130{,}000 \ (MC_{8.75,20}) = \$1{,}148.82$

$130{,}000 - 94{,}394 = (1{,}148.82 - 731.83) \ (PVAF_{i/12,240})$

$35{,}606 = 416.99 \ (PVAF_{i/12,240})$

$i = 12.994\%$

B. $Pmt_{WRAP} = 130{,}000 \ (MC_{8.75,30}) = \$1{,}022.71$

$35{,}606 = 290.88 \ (PVAF_{i/12,240}) + 1{,}022.71 \ (PVAF_{i/12,120}) \ (PVF_{i/12,240})$

$i = 11.235\%$

C. Balance of wrap at EOY20 = $81,604

$35{,}606 = 290.88 \ (PVAF_{i/12,240}) + 81{,}604 \ (PVF_{i/12,240})$

$i = 11.483\%$

D. $Pmt_{EX.\ LOAN} = 94{,}394 \ (MC_{7,12}) = \$970.73$

$Pmt_{WRAP} = 130{,}000 \ (MC_{8.75,12}) = \$1{,}461.20$

$35{,}606 = (1{,}461.20 - 970.73) \ (PVAF_{i/12,144})$

$i = 13.046\%$

E. Balance of wrap at EOY6 = $81,623

Balance of existing loan at EOY6 = $56,938

$35{,}606 = 490.47 \ (PVAF_{i/12,72}) + (81{,}623 - 56{,}938) \ (PVF_{i/12,72})$

$i = 13.145\%$

SOLUTIONS TO CHAPTER 5 FIXED-RATE MORTGAGE PROBLEMS

Some solutions show the keystrokes for the financial calculator. The calculator should be set on 12 payments per year.

5.1.

A. Payment = Amount Borrowed ($MC_{i,n}$)

$$= 125{,}000\ (MC_{8,30})$$

$$= 125{,}000\ \left[\frac{(.08/12)\ (1 + .08/12)^{360}}{(1 + .08/12)^{360} - 1} \right]$$

$$= \$917.21$$

<u>Keystrokes (Assuming calculator is set on 12 payments per year)</u>
Enter: 125,000 -PV
8 I/YR
360 N
Solve: **PMT**

B. $BAL_{EOY1} = 917.21\ (PVAF_{8/12,348})$

$$= 917.21\ \left[\frac{(1 + .08/12)^{348} - 1}{(.08/12)\ (1 + .08/12)^{348}} \right]$$

$$= \$123{,}956$$

<u>Keystrokes</u>
Enter: 917.21 PMT
8 I/YR
348 N
Solve: **PV**

5.2.

A. Pmt = 80,000 ($MC_{10,30}$) = \$702.06

Bal_{EOY5} = 77,259 - 8,000 = \$69,259

69,259 = 702.06 ($PVAF_{10/12,n}$)

n = 208 payments or 17.34 years

<u>Keystrokes</u>
Enter: 80000 -PV
10 I/YR
360 N
Solve: **PMT**
Enter: 300 N
Solve: PV (for balance)
Enter: 69259 -PV
Solve: **N**

B. Pmt = 69,259 ($MC_{10,25}$) = \$629.36

Enter: 69259 -PV
10 I/YR
300 N
Solve: **PMT**

5.3.

Pmt = 90,000 ($MC_{7.25,30}$) = $613.96

90,000 - (.02) 90,000 = 613.96 ($PVAF_{i/12,360}$)

APR = 7.45%

<u>Keystrokes</u>
-88,200 PV
613.96 PMT
360 N
I/YR

5.4.

A. 126,000 - 3,200 = $122,800

B. 122,800 = 881.01 ($PVAF_{i/12,360}$)
i = APR = 7.76%

<u>Keystrokes</u>
-122,800 PV
881.01 PMT
360 N
I/YR

C. 122,800 = 881.01 ($PVAF_{i/12,60}$) + 119,218 ($PVF_{i/12,60}$)
i = 8.13%

<u>Keystrokes</u>
-122,800 PV
881.01 PMT
119,218 FV
60 N
I/YR

D. 122,800 = 881.01 ($PVAF_{i/12,60}$) + 119,218 (1.03) ($PVF_{i/12,60}$)
i = 8.61%

<u>Keystrokes</u>
-122,800 PV
881.1 PMT
122,795 FV
60 N
I/YR

5.5.

A. 130,000 - Discount Points = 953.89 ($PVAF_{8.5/12,360}$)

Discount Points = $5,942.62 or 4.57 points

B. 130,000 - Discount Points = 953.89 ($PVAF_{8.5/12,120}$) + 114,042 ($PVF_{8.5/12,120}$)

Points = $4,175 or 3.21 points

5.6.

Loan A: 80,000 - 4,000 = 532.24 ($PVAF_{i/12,360}$) i = 7.52%

Loan B: 80,000 - 800 = 559.37 ($PVAF_{i/12,360}$) i = 7.60%

5.7.

A. Pmt = 75,000 ($MC_{9.5,30}$) = $630.64

B. Bal_{EOY1} = 630.64 ($PVAF_{9.5/12,348}$) = $74,538

C. Interest Paid in Year One:

Total Paid in Year One = 630.64 (12) = $7,568

Repayment of Principal in Year One = Bal_{EOY0} - Bal_{EOY1} = 75,000 - 74,538 = $462

Interest Paid = Total Paid - Rep. of Princ. = 7,568 - 462 = $7,105

D. Bal_{EOY6} = 630.64 ($PVAF_{9.5/12,288}$) = $71,438

E. Bal_{EOY0} - Bal_{EOY6} = 75,000 - 71,438 = $3,562 Rep. of Principal

Total Paid = 630.64 (72) = $45,406

Interest = Total Paid - Rep. of Principal = 45,406 - 3,562 = $41,844

5.8.

A. Pmt = 90,000 ($MC_{7.5,15}$) = $834.31

B. Interest in Month One = (.075/12) 90,000 = $562.50

C. Repayment of Principal in Month One = 834.31 - 562.50 = $271.81

D. Interest in Month 120 = Bal_{EOM119} (i/12) = 42,207 (.075/12) = $263.79

E. Rep. of Principal in Month 120 = 834.31 - 263.79 = $570.52

F. Interest in Month 180 = Bal_{EOM179} (i/12) = 829.13 (.075/12) = $5.18

G. Rep. of Principal in Month 180 = 834.31 - 5.18 = $829.13

5.9.

80,000 = 761.86 ($PVAF_{i/12,360}$) i = 11%

5.10.

Amt. Borrowed = 607.48 ($PVAF_{8.5/12,240}$) = $70,000

5.11.

Pmt = 90,000 (.95) (.80) ($MC_{9.5,30}$) = $575.14

5.12.

A. Option A: $.9725 = .00924\ (PVAF_{i/12,360})$ APR = 10.97%
Option B: $.9775 = .00933\ (PVAF_{i/12,360})$ APR = 11.03%
Option C: $.9850 = .00952\ (PVAF_{i/12,360})$ APR = 11.19%

B. Option A: $.9725 = .00924\ (PVAF_{i/12,60}) + .9695\ (PVF_{i/12,60})$ i = 11.36%
Option B: $.9775 = .00933\ (PVAF_{i/12,60}) + .9703\ (PVF_{i/12,60})$ i = 11.34%
Option C: $.9850 = .00952\ (PVAF_{i/12,60}) + .9716\ (PVF_{i/12,60})$ i = 11.40%

C. Option A: $.9725 = .00924\ (PVAF_{i/12,60}) + .9695\ (1.03)\ (PVF_{i/12,60})$ i = 11.80%

Option B: $.9775 = .00933\ (PVAF_{i/12,60}) + .9703\ (1.03)\ (PVF_{i/12,60})$
i = 11.78%

Option C: $.9850 = .00952\ (PVAF_{i/12,60}) + .9716\ (1.03)\ (PVF_{i/12,60})$
i = 11.83%

5.13.

A. Pmt = $100{,}000\ (MC_{12,30})$ = $1,028.61

B. Pmt = $100{,}000\ (MC_{12,15})$ = $1200.17

5.14.

Pmt = $75{,}000\ (MC_{10,15})$ = $805.95

Bal_{EOY5} = $805.95\ (PVAF_{10/12,120})$ = $60,987

5.15.

Pmt = $80{,}000\ (MC_{10,30})$ = $702.06

Bal_{EOY7} = 75,719 - 10,000 = $65,719

$65{,}719 = 702.06\ (PVAF_{10/12,n})$

n = 182.49 Payments Remaining After Partial Pay-Down

15.21 Years to Maturity

5.16.

A. 96,000 - (.03) 96,000 = 96,000 - 2880 = $93,120

B. $93{,}120 = 704.41\ (PVAF_{i/12,360})$ i = 8.32%

C. Bal_{EOY5} = 91,267 (1.02) = $93,092

$93120 = 704.41\ (PVAF_{i/12,60}) + 93{,}092\ (PVF_{i/12,60})$

i = 9.07%

5.17.

11% with 3 points:

Pmt = $100{,}000\ (MC_{11,30})$ = $952.32

Bal_{EOY8} = $952.32\ (PVAF_{11/12,264})$ = $94,549

$100{,}000 - 3{,}000 = 952.32\ (PVAF_{i/12,96}) + 94{,}549\ (PVF_{i/12,96})$

$i = 11.59\%$

10% with 5 points:

$Pmt = 100{,}000\ (MC_{10,30}) = \$877.57$

$Bal_{EOY8} = 877.57\ (PVAF_{10/12,264}) = \$93{,}533$

$100{,}000 - 5{,}000 = 877.57\ (PVAF_{i/12,96}) + 93{,}533\ (PVF_{i/12,96})$

$i = 10.96\%$

5.18.

A. Pmt = \$125,000 (.06/12) = \$625.00
B. Pmt = 125,000 ($MC_{6,20}$) = \$895.54
C. Balance EOY25 = \$46,322

5.19.

A. Pmt = 120,000 ($MC_{5,30}$) = \$644.18
B. Balance at EOY7 = \$105,534
C. Pmt = 105,534 ($MC_{6.50,23}$) = \$737.75

5.20.

A. APR = 6.44%
B. Effective Cost with 6 year holding period = 7.186%
C. 8.61 points

5.21.

A. Number of payments = 277.34
B. Net interest saved = \$52,758

5.22.

Number of payments = 214.60

5.23.

A. APR = 6.142%, Points to equalize APRs = 4.13
B. Effective cost over 5 year holding period = 6.362%, Points to equalize effective costs = 2.53

SOLUTIONS TO CHAPTER 6 PROBLEMS

6.1. i_1 = 5%
i_2 = 7% 5.5 + 2.5 = 8% or 5 + 2 = 7%, whichever is less
i_3 = 9% 7 + 2.5 = 9.5% or 7 + 2 = 9%, whichever is less
i_4 = 11% 8.75 + 2.5 = 11.25% or 9 + 2 = 11%, whichever is less

6.2. A. i_1 = 6.0 + 2.5 = 8.5%

Pmt_1 = 120,000 ($MC_{8.5,30}$) = $922.70

<u>Keystrokes</u>
-120,000 PV
8.5 I/YR
360 N
PMT

B. Bal_{EOY1} = $119,093

<u>Keystrokes</u>
922.70 PMT
8.5 I/YR
348 N
PV

i_2 = 7.5 + 2.5 = 10%

Pmt_2 = 119,093 ($MC_{10,29}$) = $1,050.97

C. 120,000 - 2400 = 922.70 ($PVAF_{i/12,12}$) + 1,050.97 ($PVAF_{i/12,12}$) ($PVF_{i/12,12}$)

+ 118,357 ($PVF_{i/12,24}$)

i = 10.33%

D. $i_{2\text{-}30}$ = 7.5 + 2.5 = 10%

$Pmt_{2\text{-}30}$ = 1,050.97

120,000 - 2,400 = 922.70 ($PVAF_{i/12,12}$) + 1,050.97 ($PVAF_{i/12,348}$) ($PVF_{i/12,12}$)

i = 10.07%

6.3. A. Pmt_1 = 100,000 ($MC_{6,30}$) = $599.55

Bal_{EOY1} = 98,772

Pmt_2 = 98,772 ($MC_{7.5,29}$) = $697.05

Bal_{EOY2} = 97,782

$Pmt_3 = 97{,}782\ (MC_{9.5,28}) = \$832.98$

$Bal_{EOY3} = 97{,}043$

$Pmt_4 = 97{,}043\ (MC_{10.5,27}) = \$902.80$

$Bal_{EOY4} = 96{,}368$

$Pmt_5 = 96{,}368\ (MC_{11.75,26}) = \$991$

$Bal_{EOY5} = 95{,}767$

B. $98{,}000 = 599.55\ (PVAF_{i/12,12}) + 697.05\ (PVAF_{i/12,12})\ (PVF_{i/12,12})$

$+ 832.98\ (PVAF_{i/12,12})\ (PVF_{i/12,24}) + 902.80\ (PVAF_{i/12,12})\ (PVF_{i/12,36})$

$+ 991(PVAF_{i/12,12})\ (PVF_{i/12,48}) + 95{,}767\ (PVF_{i/12,60})$

$i = 9.27\%$

C. $98{,}000 = 599.55\ (PVAF_{i/12,12}) + 697.05\ (PVAF_{i/12,12})\ (PVF_{i/12,12})$

$+ 832.98\ (PVAF_{i/12,12})\ (PVF_{i/12,24}) + 902.80\ (PVAF_{i/12,12})\ (PVF_{i/12,36})$

$+ 991\ (PVAF_{i/12,312})\ (PVF_{i/12,48})$

$i = 10.58\%$

6.4. A. $Pmt_1 = 85{,}000\ (MC_{6.5,30}) = \$537.26$

B. $Bal_{EOY1} = 84{,}050$

$i = 8.50\%$ 7.75 + 2.50 or 6.50 + 2, whichever is less

$Pmt_2 = 84{,}050\ (MC_{8.5,29}) = \$651.19$

C. $Bal_{EOY2} = 83{,}353$

$85{,}000 - 1{,}700 = 537.26\ (PVAF_{i/12,12}) + 651.19\ (PVAF_{i/12,12})\ (PVF_{i/12,12})$

$+ 83{,}353\ (1.03)\ (PVF_{i/12,24})$

$i = 9.91\%$

D. $Pmt_3 = 83{,}353\ (MC_{10.5,28}) = \$770.60$

$Bal_{EOY15} = 69{,}713$

$i = 10.50\%$ 8.00 + 2.50 or 8.50 + 2.00, whichever is less

$83{,}300 = 537.26\ (PVAF_{i/12,12}) + 651.19\ (PVAF_{i/12,12})\ (PVF_{i/12,12})$

$+ 770.60\ (PVAF_{i/12,156})\ (PVF_{i/12,24}) + 69{,}713\ (PVF_{i/12,180})$

$i = 10.03\%$

6.5. FRM: $100,000 - 2,500 = 933.48 (PVAF_{i/12,36})$

$+ 98,409 (PVF_{i/12,36})$

$i = 11.75\%$

ARM: $96,250 = 786.70 (PVAF_{i/12,12}) + 931.19 (PVAF_{i/12,12}) (PVF_{i/12,12})$

$+ 1,080.26 (PVAF_{i/12,12}) (PVF_{i/12,24}) + 98,360 (PVF_{i/12,36})$

$i = 12.09\%$

6.6. A. $Pmt_1 = 95,000 (MC_{5,30}) = \$509.98$

B. $Bal_{EOY1} = 93,598 (1.02) = \$95,470.37$

$Pmt_2 = 95,470.37 (MC_{5,29}) = \$520.18$

C. $Bal_{EOY2} = 93,967.60 (1.06) = \$99,605.65$

$Pmt_3 = 99,605.65 (MC_{5,28}) = \$551.39$

D. $Bal_{EOY3} = 97,931 (.97) = \$94,993$

$95,000 = 509.98 (PVAF_{i/12,12}) + 520.18 (PVAF_{i/12,12}) (PVF_{i/12,12})$

$+ 551.39 (PVAF_{i/12,12}) (PVF_{i/12,24}) + 94,993 (PVF_{i/12,36})$ $i = 6.65\%$

E. $95,000 = 509.98 (PVAF_{i/12,12}) + 520.18 (PVAF_{i/12,12}) (PVF_{i/12,12})$

$+ 551.39 (PVAF_{i/12,12}) (PVF_{i/12,24}) + 534.85 (PVAF_{i/12,324}) (PVF_{i/12,36})$

$i = 5.40\%$

6.7. A.

Year	Grad. Pmt Factor	PVAF[a]	PVF[c]	Factors[d]	Pmt
1	1.0	11.3745	1.0	11.3745	**$600.34[e]**
2	$(1.075)^1$	11.3745	.9052	11.0684	**645.36**
3	$(1.075)^2$	11.3745	.8194	10.7705	**693.77**
4	$(1.075)^3$	11.3745	.7417	10.4806	**745.80**
5	$(1.075)^4$	11.3745	.6714	10.1990	**801.74**
6-30	$(1.075)^5$	110.0472[b]	.6078	96.0225	**861.87**
Total				149.9155	

[a]Present Value Annuity Factor (PVAF) for 10%/12 for 12 periods.
[b]Present Value Annuity Factor (PVAF) for 10%/12 for 300 periods.
[c]Present Value of One Dollar (PVF) at 10%/12.
[d]Factors = Grad. Pmt. Factor x PVAF x PVF
[e]Initial payment = $90,000/149.9155 = $600.34.

B. Set up a schedule to show the amortization of the loan for the first six months.

Month	Beginning Balance	Interest	Payment	Difference
1	90,000	750	600.34	149.66
2	90,149.66	751.25	600.34	150.91
3	90,300.57	752.50	600.34	152.16
4	90,452.73	753.77	600.34	153.43
5	90,606.16	755.05	600.34	154.71
6	90,760.87	756.34	600.34	156.00
7	90,916.87			

C. Balance at the end of year 1.

149.66 ($FVAF_{10/12,12}$) = \$1,880.56

90,000 + 1,880.56 = \$91,880.56 = Bal_{EOY1}

D. 765.67 - 645.36 = 120.31 ($FAVF_{10/12,12}$) = 1511.76 + 91,880.56 = 93,392.32 = Bal_{EOY2}
778.27 - 693.77 = 84.50 ($FAVF_{10/12,12}$) = 1061.79 + 93,392.32 = 94,454.11 = Bal_{EOY3}
787.12 - 745.80 = 41.32 ($FVAF_{10/12,12}$) = 519.21 + 94,454.11 = **94,973.32** = Bal_{EOY4}
791.44 - 801.74 = -10.30 ($FVAF_{10/12,12}$) = -129.42 + 94,973.32 = 94,843.89 = Bal_{EOY5}
790.37 - 861.87 = -71.50 ($FVAF_{10/12,12}$) = -898.44 + 94,843.89 = 93,945.45 = Bal_{EOY6}

The maximum owed on the loan is \$94,973.32.

E. The payment on a standard, fixed-rate mortgage would be

90,000 ($MC_{10,30}$) = \$789.81

F. With two discount points the effective cost of the loan to the borrower is

90,000 - 1,800 = 600.34 ($PVAF_{i/12,12}$) + 645.36 ($PVAF_{i/12,12}$) ($PVF_{i/12,12}$)

+ 693.77 ($PVAF_{i./12,12}$) ($PVF_{i/12,24}$) + 745.80 ($PVAF_{i/12,12}$) ($PVF_{i/12,36}$)

+ 801.74 ($PVAF_{i/12,12}$) ($PVF_{i/12,48}$) + 861.87 ($PVAF_{i/12,300}$) ($PVF_{i/12,60}$)

i = 10.225%

G. The effective cost of a standard, fixed-rate mortgage would be

90,000 - 1,800 = 789.81 ($PVAF_{i/12,360}$) i = 10.24%

6.8. A. Pmt = 120,000 ($MC_{7,30}$) = \$798.36

B. 135,000 $(1.06)^5$ (.50) = (180,660 - 135,000) (.50) = 45,660 (.50) = \$22,830

C. 120,000 = 798.36 ($PVAF_{i/12,60}$) + (112,958 + 22,830) ($PVF_{i/12,60}$) i = 10.02%

D. 120,000 = 798.36 ($PVAF_{11/12,60}$) + (112,958 + X) ($PVF_{11/12,60}$)

(112,958 + X) must equal \$143,986 thus lender's share must equal \$31,027

Total appreciation would be \$62,056 raising the price to \$197,056 at the end of year five. This

would require an annual appreciation rate of 7.858%.

E. $(135{,}000\ (1.06)^3 - 135{,}000)/2 = \$12{,}894$

$120{,}000 = 798.36\ (PVAF_{i/12,36}) + (116{,}072 + 12{,}894)\ (PVF_{i/12,36})$ $\qquad i = 10.125\%$

6.9.

Year	Beginning Balance	Payment	Interest	Ending Balance
1	0	17,045.65		17,045.65
2	17,045.65	17,045.65	1,363.65	35,454.95
3	35,454.94	17,045.65	2,836.40	55,336.98
4	55,336.98	17,045.65	4,426.96	76,809.57
5	76,809.57	17,045.65	6,144.77	100,000
Total Interest Paid			14,771.78	

6.10. A. Refinancing the payoff for the remaining term

$Pmt_{Ex} = 120{,}000\ (MC_{9.5,30}) = \$1009.02$

$Bal^{Ex}_{EOY5} = \$115{,}489$

Payoff of existing loan = outstanding balance + prepayment penalty = 115,489 + 4,620 = $120,109

Financing Costs of new loan = 120,109 (0.06) = $7,207

$Pmt_{new} = 120{,}109\ (MC_{7.25,25}) = \$868.16$

$NPV = 140.86\ (PVAIF_{7.25/12,300}) - 7{,}207 = \$12{,}281$

B. Refinancing the payoff for 30 years

Payoff of existing loan = $120,109

Financing costs of new loan = $7,207

$Pmt_{new} = 120{,}109\ (MC_{7.25,30}) = \$819.36$

$NPV = 189.66\ (PVAIF_{7.25/12,300}) - 819.36\ (PVAIF_{7.25/12,60})\ (PVIF_{7.25/12,300}) - 7{,}207$

$NPV = \$12{,}280$

C. Refinancing the payoff for 15 years

$Pmt_{new} = 120{,}109\ (MC_{7.25,15}) = \$1{,}096.43$

$NPV = -87.41\ (PVAIF_{7.25/12,180}) + 1009.02\ (PVAIF_{7.25/12,120})\ (PVIF_{7.25/12,180}) - 7{,}207$

$NPV = \$12{,}282$

D. Refinancing for the remaining term with no out-of-pocket expenses

Amount of new loan = payoff of existing loan + financing costs of new loan

127,776 = 120,109 / .94 (financing costs are 6% of loan amount)

Pmt_{new} = 127,776 ($MC_{7.25,25}$) = $923.57

NPV = 85.45 ($PVAIF_{7.25/12,300}$) - 0 = $11,822

E. Extending the term of the new loan with no out-of-pocket expenses

Pmt_{new} = 127,776 ($MC_{7.25,30}$) = $871.66

NPV = 137.36 ($PVAIF_{7.25/12,300}$) - 871.66 ($PVAIF_{7.25/12,60}$) ($PVIF_{7.25/12,300}$)

NPV = $11,822

F. Assume the conditions in Part A except the borrower will hold the mortgage for only six more years

With no refinancing, the payoff on the existing loan at the end of year 11 is the balance at the end of year 11 plus the prepayment penalty = 106,343 + 4,254 = $110,597

With refinancing, the new loan will be six years old and the payoff is the outstanding balance at the end of year 6 = $107,303

NPV = 140.86 ($PVAIF_{7.25/12,72}$) + 3,294 ($PVIF_{7.25/12,72}$) - 7,207

NPV = $3,132

G. Refinancing for 25 years and taking out equity of $40,000 with no out-of-pocket expenses

New loan amount = (120,109 + 40,000) / .94 = $170,329

Pmt_{new} = 170,329 ($MC_{7.25,25}$) = $1,231.15

NPV = -222.13 ($PVAIF_{7.25/12,300}$) + 40,000

NPV = $9,268

6.11. A. Pmt = $140,000 (.045/12) = $525.00
B. Pmt = $140,000 ($MC_{6,27}$) = $873.58
C. Balance EOY4 = $137,859

6.12. A. Pmt in year one = $853.29
B. Pmt in year two = $987.76
C. APR = 8.01%

6.13. A. Pmt in year one = $709.74
B. Pmt in year two = $870.27
C. Effective Cost = 9.06%

6.14. A. Pmt in interest-only period = $458.33
B. Pmt to fully amortize the debt = $721.11
C. 9.20 points

6.15. A. Pmt in year one = $738.13
B. Pmt in year two = $760.27
C. Effective Cost = 5.076%

6.16. A. NPV = $10,127
B. NPV = $10,127
C. NPV = $9,441
D. NPV = -$4,907
E. NPV = $7,526

SOLUTIONS TO CHAPTER 8 PROBLEMS

8.1. $125{,}000 - 3{,}750 = 874.02\ (PVAF_{i/12,360})$

$i = 7.814\%$, thus the APR $= 7.875\%$

8.2. $90{,}000 - 1{,}800 = 483.14\ (PVAF_{i/12,12}) + 566.80\ (PVAF_{i/12,12})\ (PVF_{i/12,12})$

$+ 674.49\ (PVAF_{i/12,336})\ (PVF_{i/12,24})$

$i = 8.04\%$, thus the APR $= 8.00\%$

8.3. $140{,}000 - 3{,}500 = 839.37\ (PVAF_{i/12,12}) + 931.42\ (PVAF_{i/12,12})\ (PVF_{i/12,12})$

$+ 1027.27\ (PVAF_{i/12,336})\ (PVF_{i/12,24})$

$i = 8.024\%$, thus the APR $= 8\%$

SOLUTIONS TO CHAPTER 14 PROBLEMS

14.1.

A. interest = $11,660,000 (106,000,000 x .11), principal reduction = $532,607 (12,192,607 - 11,660,000)
B. $1,371,076 (.013 x (106,000,000 - 532,607)
C. $104,096,316 (106,000,000- 532,607 - 1,371,076)
D. $13,563,684 (12,192,607 +1,371,076)

14.2.

A. $2,742,152 (.026 x (106,000,000 - 532,607)
B. $102,725,240 (106,000,000 -532,607 - 2,742,152)
C. $14,934,760 (12,192,607 + 2,742,152)

14.3.

Year	Pool Balance	Scheduled P&I	Prepayments	Amortization	Interest	Amount Owed Security Holders
0	106,000,000					$100,000,000
1	102,486,566	10,317,653	2,735,781	777,653	9,540,000	96,486,566
2	94,138,061	10,049,394	7,522,901	825,603	9,223,791	88,138,061
3	82,574,701	9,305,740	10,730,046	833,315	8,472,426	76,574,701
4	71,958,342	8,235,580	9,812,502	803,857	7,431,723	65,958,342
5	62,644,808	7,247,310	8,542,475	771,060	6,476,251	56,644,808

14.4.

[(108-104)/104] x 1.09/.01 = .038462 x 109 = 4.192

14.5.

[(109-104)/104] x 1.09/.01 = .048077 x 109 = 5.24

14.6.

$758.34 (FV= 1,000, PMT= 90, N= 30, I/YR = 12)

14.7.

$826.12 (same calculator entries except I/YR = 11)

14.8.

D = $\Delta p/p$ x (1+r)/Δr = 67.78/758.34 x 1.12/.01 = .0894 x 112 = 10.01 years.

14.9.

Mortgage Passthrough.

Part 1

Year	Ending Balance	Interest	Principal	Prepayment	Total Payment
1	109,193,001	9,900,000	806,999	0	10,706,999

2	108,313,373	9,827,370	879,629	0	"
3	107,354,578	9,748,204	958,795	0	"
4	106,309,491	9,661,912	1,045,087	0	"
5	105,170,347	9,567,854	1,139,144	0	"
	PV @ 9%	37944662	72055339		

Part 2

Year	Ending Balance w/ Prep	Ending Balance w/o Prep	Interest	Principal
1	107,773,492	109,193,001	9,900,000	806,999
2	102,949,803	106,905,299	9,699,614	868,193
3	96,171,273	102,038,486	9,265,482	911,316
4	89,520,952	95,235,055	8,655,415	936,218
5	83,248,000	88,561,702	8,056,886	959,249
	PV @ 9%		35,769,329	3,461,496

Year	Prepayment	Regular Payment	Total Payment
1	1,419,509	10,706,999	12,126,508
2	3,955,496	10,567,808	14,523,304
3	5,867,213	10,176,799	16,044,012
4	5,714,103	9,591,633	15,305,736
5	5,313,702	9,016,135	14,329,837
	16,663,685		

14.10.
Part 1

Year	Ending Balance w/o Prep.	Principal	Interest	Prepayment	Normal Payment
1	107,046,637	953,363	8,640,000	0	9,593,363
2	106,017,005	1,029,632	8,563,731	0	9,593,363
3	104,905,003	1,112,002	8,481,360	0	9,593,363
4	103,704,040	1,200,963	8,392,400	0	9,593,363
5	102,407,001	1,297,040	8,296,323	0	9,593,363
					+ 102,407,001

Year	Total Cash Inflow	Bond Payment	Fund Contribution	Fund Balance
1	9,593,363	7,000,000	2,593,363	2,593,363
2	9,593,363	7,000,000	2,593,363	5,342,328
3	9,593,363	7,000,000	2,593,363	8,256,630
4	9,593,363	7,000,000	2,593,363	11,344,967
5	9,593,363	7,000,000	2,593,363	14,619,028
	+ 102,407,001	+ 100,000,000		+ 2,407,001

NPV of residual @ 8% = $3,587,629

Part 2

Year	Ending Balance w/ Prep	Ending Balance w/o Prep	Principal	Interest	Normal Payment
1	105,655,031	107,046,637	953,363	8,640,000	9,593,363
2	100,767,149	104,638,784	1,016,247	8,452,402	9,468,649
3	93,976,875	99,710,212	1,056,937	8,061,372	9,118,309
4	87,326,957	92,901,018	1,075,856	7,518,150	8,594,006
5	81,060,663	86,234,748	1,092,209	6,986,157	8,078,366

Year	Prepayment	Total Cash Inflow	Bond Payment	Fund Contribution
1	1,391,606	10,984,969	7,000,000	3,984,969
2	3,871,635	13,340,284	7,000,000	6,340,284
3	5,733,337	14,851,646	7,000,000	7,851,646
4	5,574,061	14,168,067	7,000,000	7,168,067
5	5,174,085	13,252,451	7,000,000	6,252,451
		+ 81,060,663	+ 100,000,000	

Year	Fund Balance
1	3,984,969
2	10,564,351
3	19,049,858
4	27,360,917
5	35,255,023
	- 18,939,337

NPV of residual @ 8% = $3,104,182

SOLUTIONS TO CHAPTERS 15-16 PROBLEMS

15/16.1.

AFTER-TAX CASH FLOW FROM OPERATIONS

	Year 1	2	3	4	5
GR	273000	286650	300982	316032	331833
- VAC	10920	11466	12039	12641	13273
+ OI	10080	10584	11113	11669	12252
EGI	272160	285768	300057	315060	330812
- OE	122472	128596	135026	141777	148865
NOI	149688	157172	165031	173283	181947
- MP[a]	84563	84563	84563	84563	108050
BTCF	65125	72609	80468	88720	73897
- TXS	15528	17706	20234	22915	16703
ATCF	49597	54903	60234	65805	57194

[a] MP is the monthly mortgage payment multiplied by twelve.

TAXES FROM OPERATIONS

	Year 1	2	3	4	5
EGI					
- OE					
NOI	149688	157172	165031	173283	181947
- INT	76358	75317	74145	72824	94822
- AFC	640	640	640	640	10240
- DEP	17231	17980	17980	17980	17231
+ RR	0	0	0	0	0
TI	55459	63235	72266	81839	59654
x t	.28	.28	.28	.28	.28
TXS	15528	17706	20234	22915	16703

AFTER-TAX CASH FLOW FROM RESALE OF PROPERTY

ESP	1021025
- SE	81682
NSP	939343
- UMB	587164
BTER	352179
- TXR	39252
ATER	312927

TAXES FROM RESALE OF PROPERTY

ESP	
- SE	
AR	939343
- AB	736598
TG	202745
DR	88402
x t_d	.25
DRT	22101
CG	114343
x t_g	.15
CGT	17151
DRT	22101
+ CGT	17151
TXR	39252

SUMMARY OF AFTER-TAX CASH FLOWS

	Year					
	0	1	2	3	4	5
ATCF	-197800	49597	54903	60234	65805	57194
ATER						312927

NPV @ 12% = $184,962
IRR = 34.18%

15/16.2.

AFTER-TAX CASH FLOW FROM OPERATIONS

	Year 1	2	3	4	5
GR	322500	338625	355556	373334	392001
- VAC	12900	13545	14222	14933	15680
+ OI	0	0	0	0	0
EGI	309600	325080	341334	358401	376321
- OE	130032	136534	143360	150528	158055
NOI	179568	188546	197974	207873	218266
- MP[a]	83378	83378	83378	83378	109241
BTCF	96190	105168	114596	124495	109025
- TXS	24949	27589	30615	33813	27151
ATCF	71241	77579	83981	90682	81874

[a] MP is the monthly mortgage payment multiplied by twelve.

TAXES FROM OPERATIONS

	Year 1	2	3	4	5
EGI					
- OE					
NOI	179568	188546	197974	207873	218266
- INT	71464	70216	68838	67315	91497
- AFC	720	720	720	720	11520
- DEP	18282	19077	19077	19077	18282
+ RR	0	0	0	0	0
TI	89102	98533	109339	120761	96967
x t	.28	.28	.28	.28	.28
TXS	24949	27589	30615	33813	27151

AFTER-TAX CASH FLOW FROM RESALE OF PROPERTY

ESP	1094988
- SE	87599
NSP	1007389
- UMB	646578
BTER	360811
- TXR	35057
ATER	325754

TAXES FROM RESALE OF PROPERTY

ESP	
- SE	
AR	1007389
- AB	836205
TG	171184
DR	93795
x t_d	.25
DRT	23449
CG	77389
x t_g	.15
CGT	11608
DRT	23449
+ CGT	11608
TXR	35057

SUMMARY OF AFTER-TAX CASH FLOWS

	Year 0	1	2	3	4	5
ATCF	-224400	71241	77579	83981	90682	81874
ATER						325754

NPV @ 15% = $205,939
IRR = 39.22%

15/16.3. A.

AFTER-TAX CASH FLOW FROM OPERATIONS

	Year 1	2	3	4	5
GR	15300	16218	17191	18222	19316
- VAC	1275	1352	1433	1519	1610
+ OI	0	0	0	0	0
EGI	14025	14866	15758	16703	17706
- OE	4488	4847	5235	5654	6106
NOI	9537	10019	10523	11050	11600
- MP[a]	11728	11728	21728	10528	15160
BTCF	- 2191	- 1709	-11205	522	- 3560
- TXS	+1268	+1134	+ 948	+ 415	+2174
ATCF	- 923	- 575	-10257	937	- 1386

[a] MP is the monthly mortgage payment multiplied by twelve.

TAXES FROM OPERATIONS

	Year 1	2	3	4	5
EGI					
- OE					
NOI	9537	10019	10523	11050	11600
- INT	10488	10345	10185	8807	13239
- AFC	170	170	170	170	2720
- DEP	3406	3555	3555	3555	3406
+ RR	0	0	0	0	0
TI	- 4527	- 4051	- 3387	- 1482	- 7765
x t	.28	.28	.28	.28	.28
TXS	- 1268	- 1134	- 948	- 415	- 2174

AFTER-TAX CASH FLOW FROM RESALE OF PROPERTY

ESP	142000
- SE	11360
NSP	130640
- UMB	77192
BTER	53448
- TXR	6715
ATER	46733

TAXES FROM RESALE OF PROPERTY

ESP	
- SE	
AR	130640
- AB	97523
TG	33117
DR	17477
x t_d	.25
DRT	4369
CG	15640
x t_g	.15
CGT	2346
DRT	4369
+ CGT	2346
TXR	6715

SUMMARY OF AFTER-TAX CASH FLOWS

	Year 0	1	2	3	4	5
ATCF	-23400	-923	-575	-10257	937	-1386
ATER						46733

NPV @ 12% = -$5,657
IRR = 6.93%

15/16.3. B.

AFTER-TAX CASH FLOW FROM OPERATIONS

	Year 1	2	3	4	5
GR	15300	16218	17191	18222	19316
- VAC	1275	1352	1433	1519	1610
+ OI	0	0	0	0	0
EGI	14025	14866	15758	16703	17706
- OE	4488	4847	5235	5654	6106
NOI	9537	10019	10523	11050	11600
- MP[a]	9893	11331	12426	12068	12068
BTCF	- 356	- 1312	- 1903	- 1018	- 468
- TXS	+ 570	+ 953	+ 1146	+ 818	+1259
ATCF	214	- 359	- 757	- 200	791

[a] MP is the monthly mortgage payment multiplied by twelve.

TAXES FROM OPERATIONS

	Year 1	2	3	4	5
EGI					
- OE					
NOI	9537	10019	10523	11050	11600
- INT	8002	9700	10895	10249	10030
- AFC	166	166	166	166	2661
- DEP	3406	3555	3555	3555	3406
+ RR	0	0	0	0	0
TI	- 2037	- 3402	- 4093	- 2920	- 4497
x t	.28	.28	.28	.28	.28
TXS	- 570	- 953	- 1146	- 818	- 1259

AFTER-TAX CASH FLOW FROM RESALE OF PROPERTY

ESP	142000
- SE	11360
NSP	130640
- UMB	86090
BTER	44550
- TXR	6715
ATER	37835

TAXES FROM RESALE OF PROPERTY

ESP	
- SE	
AR	130640
- AB	97523
TG	33117
DR	17477
x t_d	.25
DRT	4369
CG	15640
x t_g	.15
CGT	2346
DRT	4369
+ CGT	2346
TXR	6715

SUMMARY OF AFTER-TAX CASH FLOWS

	Year					
	0	1	2	3	4	5
ATCF	-23325	214	-359	-757	-200	791
ATER						37835

NPV @ 12% = -$2,179
IRR = 9.85%

15/16.4.

AFTER-TAX CASH FLOW FROM OPERATIONS

	Year 1	Year 2	Year 3
GR	31200	32760	34398
- VAC	- 2600	- 2730	- 2867
+ OI	+ 624	+ 655	+ 688
EGI	29224	30685	32219
- OE	- 8767	- 9293	- 9851
NOI	20457	21392	22368
- MP[a]	-18562	-18562	-24708
BTCF	1895	2830	- 2340
- TXS	+ 326	+ 57	+ 3306
ATCF	2221	2887	966

[a] MP is the monthly mortgage payment multiplied by twelve.

TAXES FROM OPERATIONS

	Year 1	Year 2	Year 3
EGI			
- OE			
NOI	20457	21392	22368
- INT	-15074	-14784	-20617
- AFC	- 413	- 413	- 7424
- DEP	- 6133	- 6400	- 6133
+ RR	+ 0	+ 0	+ 0
TI	- 1163	- 205	-11806
x t	x .28	x .28	x .28
TXS	- 326	- 57	- 3306

AFTER-TAX CASH FLOW FROM RESALE OF PROPERTY

ESP	262024
- SE	- 20962
NSP	241062
- UMB	-173645
BTER	67417
- TXR	- 7826
ATER	59591

TAXES FROM RESALE OF PROPERTY

ESP	262024
- SE	- 20962
AR	241062
- AB	-201334
TG	39728

DR	18666
x t_d	x .25
DRT	4667

CG	21062
x t_g	x .15
CGT	3159

DRT	4667
+ CGT	+3159
TXR	7826

SUMMARY OF AFTER-TAX CASH FLOWS

Year

	0	1	2	3
ATCF	-43250	2221	2887	966
ATER				59591

NPV @ 12% = $4138
IRR = 15.63%

15/16.5.

After-tax Cash Flows From Installment Sale Agreement

A. ATCF IN YEAR OF SALE

Down Payment	60,000
- Selling Expenses	-5,000
- Taxes	-5720
ATCF	49,280

B. TAX IN YEAR OF SALE

Down Payment	60,000
+ Excess of Mortgage over Adj. Basis and Selling Exp.	+ 0
Total Pmt in Year of Sale	60,000
x Profit Percentage	x .4333
Taxable Portion of Gain	25,998

Depreciation Recovery	18,199
x Tax Rate	x .25
Taxes on Dep. Recovery	4,550

Capital Gain	7799
x Tax Rate	x .15
Taxes on Capital Gain	1,170

Total Taxes: 4,550 + 1,170 = 5,720

C. Profit Percentage

Sale Price	150,000
- Selling Expenses	- 5,000
- Adjusted Basis	-80,000
Total Capital Gain	65,000

Sale Price	150,000
- Mortgage Balance Assumed	- 0
+ Excess of Mortgage over Adj. Basis and Selling Expenses	+ 0
Contract Price	150,000

Total Gain/Contract Price = Profit Percentage 65,000/150,000 = .4333

D. ATCF FROM INSTALLMENTS

	1	2
Debt Service Pmt	23,742	23,742
+ Balloon Payment	+ 0	59,042
- Tax on Installments	-3,925	- 9,281
ATCF	19,817	73,503

E. TAXES ON INSTALLMENTS

Rep. of Principal	14,742	16,216
+ Balloon Payment	+ 0	+59,042
Principal Portion	14,742	75,258
x Profit Percentage	x .4333	x .4333
Taxable Principal	6,388	32,609
Dep. Recovery	4,472	22,826
x Tax Rate	x .25	x .25
Taxes on Dep. Rec.	1,118	5,707
Capital Gain	1,916	9,783
x Tax Rate	x 15	x .15
Taxes on Cap. Gain	287	1,467
Interest Earned	9,000	7,526
x Tax Rate	x .28	x .28
Tax on Interest	2,520	2,107
Tax on Principal	1,405	7,174
+ Tax on Interest	+2,520	+2,107
Tax on Installments	3,925	9,281

The depreciation recapture is 70% of the total gain and is prorated equally across the installment period.

15/16.6.

A. ATCF IN YEAR OF SALE

Down Payment	66,000
- Selling Expenses	-6,000
- Taxes	-7,310
ATCF	52,690

B. TAX IN YEAR OF SALE

Down Payment		66,000
+ Excess of Assumed Mortgage over Adj. Basis and Selling Exp.		+ 0
Total Pmt in Year of Sale		66,000
x Profit Percentage		x .5128
Taxable Portion of Gain		33,845
Depreciation Recovery	22,338	
x Tax Rate	x .25	
Taxes on Dep. Recovery	5,584	
Capital Gains	11,507	
x Tax Rate	x .15	
Taxes on Capital Gain	1,726	

Total Taxes: 5,584 + 1,726 = 7,310

C. Profit Percentage

Sale Price	220,000
- Selling Expenses	- 6,000
- Adjusted Basis	-114,000
Total Gain	100,000

Sale Price	220,000
- Balance of Assumed Mortgage	- 25,000
+ Excess of Assumed Mortgage over Adj. Basis and Selling Expenses	+ 0
Contract Price	195,000

Profit %age = Total Gain/Contract Price
.5128 = 100,000/195,000

D. ATCF FROM INSTALLMENTS

	Year 1	Year 2
Debt Service Payment	27,905	27,905
+ Balloon Payment	+ 0	+ 92,424
- Tax on Installments	- 4,838	- 14,838
ATCF	23,067	105,491

E. TAXES ON INSTALLMENTS

Rep. of Principal	17,585	18,991
+ Balloon Payment	+ 0	+ 92,424
Principal Portion	17,585	111,415
x Profit Percentage	x .5128	x .5128
Taxable Principal	9,018	57,136
Dep. Recovery	5,952	37,710
x Tax Rate	x .25	x .25
Taxes on Dep. Rec.	1,488	9,428
Capital Gains	3,066	19,426
x Tax Rate	x 15	x .15
Taxes on Cap. Gain	460	2,914
Interest Earned	10,320	8,913
x Tax Rate	x .28	x .28
Tax on Interest	2,890	2,496
Tax on Dep. Rec.	1,488	9,428
+ Tax on Capital Gain	+ 460	+ 2,914
+ Tax on Interest	+2,890	+ 2,496
Tax on Installments	4,838	14,838

15/16.7.

Tax-Deferred Exchange **Calculations**

	Larry	Bob
A. Equities Must Balance		
Market Value of Property Given	$300,000	$450,000
- Mortgage Balance	140,000	225,000
= Equity Given	160,000	225,000
+ Boot (Unlike Property)	65,000	0
= Total	225,000	225,000
B. Realized Gain		
Market Value of Property Given	300,000	450,000
- Selling Expenses	10,000	16,000
- Adjusted Basis	160,000	300,000
= Realized Gain	130,000	134,000
C. Recognized Gain		
Net Mortgage Relief	0	85,000
- Boot Given	0	65,000
- Selling Expenses	10,000	16,000
+ Boot Received	65,000	0
= Net Boot Received	55,000	4,000
Recognized Gain (Lesser of Realized Gain or Net Boot Received)	55,000	4,000
D. Tax Basis of Property Received		
Adjusted Basis	160,000	300,000
- Boot Received	65,000	0
- Old Mortgage	140,000	225,000
+ Boot Given	0	65,000
+ Selling Expenses	10,000	16,000
+ Mort. on New Property	225,000	140,000
+ Recognized Gain	55,000	4,000
= Tax Basis of New Prop.	245,000	300,000

15/16.8. **Incremental Cost of Financing**

Payment on first loan:		Payment of second loan:	
$390,000	-PV	$480,000	-PV
6/12	I/YR	6.75/12	I/YR
240	N	240	N
PMT = $2,794.08		PMT = $3,649.75	

Incremental cost of the additional $90,000:

$90,000	-PV
$855.67	PMT
240	N

Solve for I/YR = 9.78%

15/16.9. **Partitioning the IRR**

IRR = 19.90%

PV of CF from Operations: $151,887

PV of CF from Resale: $348,092

Total PV of cash flows: $151,887 + $348,092 = $499,979

CF from Operations is 30.40% of total cash flows

CF from Resale is 69.60% of total cash flows

15/16.10. **Optimal Capital Structure**

Value of Debt	Value of Equity	Total Value
0	1,218,230	1,218,230
843,750	599,231	1,442,981

Optimal Capital Structure—$843,750 Debt.

SOLUTION TO CHAPTER 18 PROBLEM

18.1.

Draw Schedule for Developer

Month	Cash to Developer	Interest	Draw	Cumulative Balance
Closing	12,500	14,250	26,750	26,750
1	36,000	268	36,268	63,018
2	36,000	630	36,630	99,648
3	36,000	996	36,996	136,644
4	36,000	1,366	37,366	174,010
5	36,000	1,740	37,740	211,750
6	36,000	2,118	38,118	249,868
7	36,000	2,499	38,499	288,367
8	36,000	2,884	38,884	327,251
9	36,000	3,273	39,273	366,524
10	36,000	3,665	39,665	406,189
11	36,000	4,062	40,062	446,251
12	36,000	4,463	40,463	486,714
	444,500	42,214	486,714	

Before-Tax Yield to the Lender

Month	Net Outflow	Inflow
Closing	12,500	---
1	36,000	---
2	36,000	---
3	36,000	---
4	36,000	---
5	36,000	---
6	36,000	---
7	36,000	---
8	36,000	---
9	36,000	---
10	36,000	---
11	36,000	---
12	36,000	486,714

Effective yield for the lender = IRR = 18.98%

SOLUTIONS TO CHAPTER 19 PROBLEMS

19.1. A.

Before-tax Cash Flows

	Year 1	Year 2	Year 3	Year 4	Year 5
GR	252,000	264,600	277,830	291,722	306,308
- VAC	20,160	21,168	22,226	23,337	24,504
+ OI	20,160	21,168	22,226	23,337	24,504
EGI	252,000	264,600	277,830	291,722	306,308
- OE	113,400	119,070	125,024	131,275	137,838
NOI	138,600	145,530	152,806	160,447	168,470
- MP	70,110	70,110	70,110	70,110	70,110
Cash flow before participation	68,490	75,420	82,696	90,337	98,360
- Participation	7,720[a]	9,106	10,561	12,089	13,694
BTCF	60,770	66,314	72,135	78,248	84,666

After-tax Cash Flows

	Year 1	Year 2	Year 3	Year 4	Year 5
NOI	138,600	145,530	152,806	160,447	168,470
- DEP	16,709	17,436	17,436	17,436	16,709
- INT	57,600	56,474	55,247	53,909	52,451
- Participation	7,720[a]	9,106	10,561	12,089	13,694
Taxable inc.	56,571	62,514	69,562	77,013	85,616
Tax due	15,840	17,504	19,477	21,564	23,972
ATCF	40,731	45,010	50,085	55,449	61,644

a[$138,600 – $100,000] x .20

Sale of Property: Year 5

Sale price	1,021,025
- Selling expenses	81,682
- Mortgage balance	565,133
- Participation	26,523[b]
- Tax	38,359[c]
ATER	309,328

[b] .12 x ($1,021,025 – $800,000)

[c] .1932% x 198,546 ($1,021,025 – $81,682 – $26,523 – $714,274). Depreciation recapture is 43.20 percent of total gain.

19.1. B.

	Year 1	Year 2	Year 3	Year 4	Year 5
Owner					
Cash flows	40,731	45,010	50,085	55,449	365,333

Initial Outlay = $172,800
Discount rate = 16%
Net present value = $135,099
IRR = 35.41%

	Year 1	Year 2	Year 3	Year 4	Year 5
Lender					
Cash flows	77,830[a]	79,216	80,671	82,199	675,460[b]

Initial Outlay = $627,200 (640,000 - 12,800)
IRR = 11.93%

[a] 70,110 + 7720
[b] 70,110 + 13,694 + 565,133 + 26,523

19.2. A.

After-Tax Cash Flow from Owning the Property

	Year					
	0	1	2	3	4	5
NOI	--	0	0	0	0	0
- MP	--	- 29925	- 29925	- 29925	- 29925	- 29925
= BTCF	--	- 29925	- 29925	- 29925	- 29925	- 29925
- TAXES	--	+ 10691	+ 10791	+ 10791	+ 10791	+ 10691
= ATCF	- $105,000	- 19234	- 19134	- 19134	- 19134	- 19234
ATER						213452

NPV @ 15% - $63,153

ATER

SP	560,000
- MB	- 315,000
- TX	- 31,548
= ATER	213,452

Tax on Resale

SP	560,000
- AB	- 377,643
= GAIN	= 182,357
x TX RATE	x .173
= TX	= 31,548

TX RATE = (.23)(.25) + (.77)(.15) = .173 (Depreciation recovery is 23 percent of total gain)

After-Tax Cash Flow from Leasing

		Year				
	0	1	2	3	4	5
NOI	--	0	0	0	0	0
- LEASE	--	- 31000	- 31000	- 31000	- 31000	- 31000
- TAXES	--	+ 8680	+ 8680	+ 8680	+ 8680	+ 8680
= ATCF	0	- 22320	- 22320	- 22320	- 22320	- 22320

NPV @ 15% - $74,820

After-tax cost of lease = 31,000 (1-t) = 31,000 (.72) = $22,320

Owning is the better option since it has the lower present value of cost.

B. The breakeven lease payment would be one that has a present value of $63,153. The lease payment would be $26,167.

C. The cost of the leasing is $74,820 thus owning is better with a present value of cost of $63,153.

After-Tax Cash Flow from Leasing

		Year				
	0	1	2	3	4	5
NOI	--	0	0	0	0	0
- LEASE	- 27612	- 27612	- 27612	- 27612	- 27612	- 0
- TAXES	+ 7731	+ 7731	+ 7731	+ 7731	+ 7731	+ 0
= ATCF	- 19881	- 19881	- 19881	- 19881	- 19881	- 0

NPV @ 15% - $76,641

19.3. A.

Retention of Property

Before-tax Cash Flows

	Year 1	Year 2	Year 3	Year 4	Year 5
NOI	226,000	226,000	226,000	226,000	226,000
MP	10,578	10,578	10,578	10,578	10,578
BTCF	215,422	215,422	215,422	215,422	215,422

After-tax Cash Flows

NOI	226,000	226,000	226,000	226,000	226,000
Interest	6,500	6,092	5,644	5,150	4,607
Depreciation	25,000	25,000	25,000	25,000	25,000
Taxable income	194,500	194,908	195,356	195,850	196,393
Tax	54,460	54,574	54,700	54,838	54,990

ATCF	160,962	160,848	160,722	160,584	160,432

B. **Sale-leaseback Cash Flows**

	Year 0	Year 1	Year 2	Year 3	Year 4	Year5
NOI		226,000	226,000	226,000	226,000	226,000
Lease exp.	0	150,000	150,000	150,000	150,000	150,000
Tax. Inc.	76,000	76,000	76,000	76,000	76,000	
Tax	0	21,280	21,280	21,280	21,280	21,280
CF	1,200,000	54,720	54,720	54,720	54,720	(1,245,280)

C. **Net Present Value: Own vs. Lease[a]**

Own

Year 1	Year 2	Year 3	Year 4	Year 5
$160,962	$160,848	$160,722	$160,584	$160,432

NPV = $551,847

Lease

Year 0	Year 1	Year 2	Year 3	Year 4	Year 5
1,200,000	54,720	54,720	54,720	54,720	(1,245,280)

NPV = $712,678

[a] Cash flow discount rate 14%

The sale-leaseback arrangement has the greater value.

19.4. A.

Year	1	2	3	4	5
Gross Rent	$525,000	$567,000	$612,360	$661.349	$714,257
Vacancy	$26,250	$28,350	$30,618	$33,067	$35,713
Net Rent	$498,750	$538,650	$581,742	$628,282	$678,544
Operating Expenses	$84,000	$89,880	$96,172	$102,904	$110,107
Net Operating Inc.	$414,750	$448,770	$485,570	$525,378	$568,437
Mortgage Payment	$183,025	$183,025	$183,025	$183,025	$183,025
Participation Pmt.	$29,655	$35,779	$42,403	$49,568	$57,319
Before Tax Cash Flow	$202,070	$229,966	$260,142	$292,785	$328,093
Net Operating Inc.	$414,750	$448,770	$485,570	$525,378	$568,437
Depreciation	$87,121	$90,909	$90,909	$90,909	$90,909
Interest	$171,000	$169,858	$168,607	$167,237	$165,737
Participation Pmt.	$29,655	$35,779	$42,403	$49,568	$57,319
Taxable Income	$126,974	$152,224	$183,651	$217,664	$254,472
Tax Due	$35,553	$42,623	$51,422	$60,946	$71,252
Before Tax Cash Flow	$202,070	$229,966	$260,142	$292,785	$328,093
Less Tax Due	$35,553	$42,623	$51,422	$60,946	$71,252
After-Tax Cash Flow	$166,517	$187,343	$208,720	$231,839	$256,841

19.4. B.

YEAR	1	2	3	4
Cash Flows to Lender	$212,680	$218,804	$225,428	$232,593